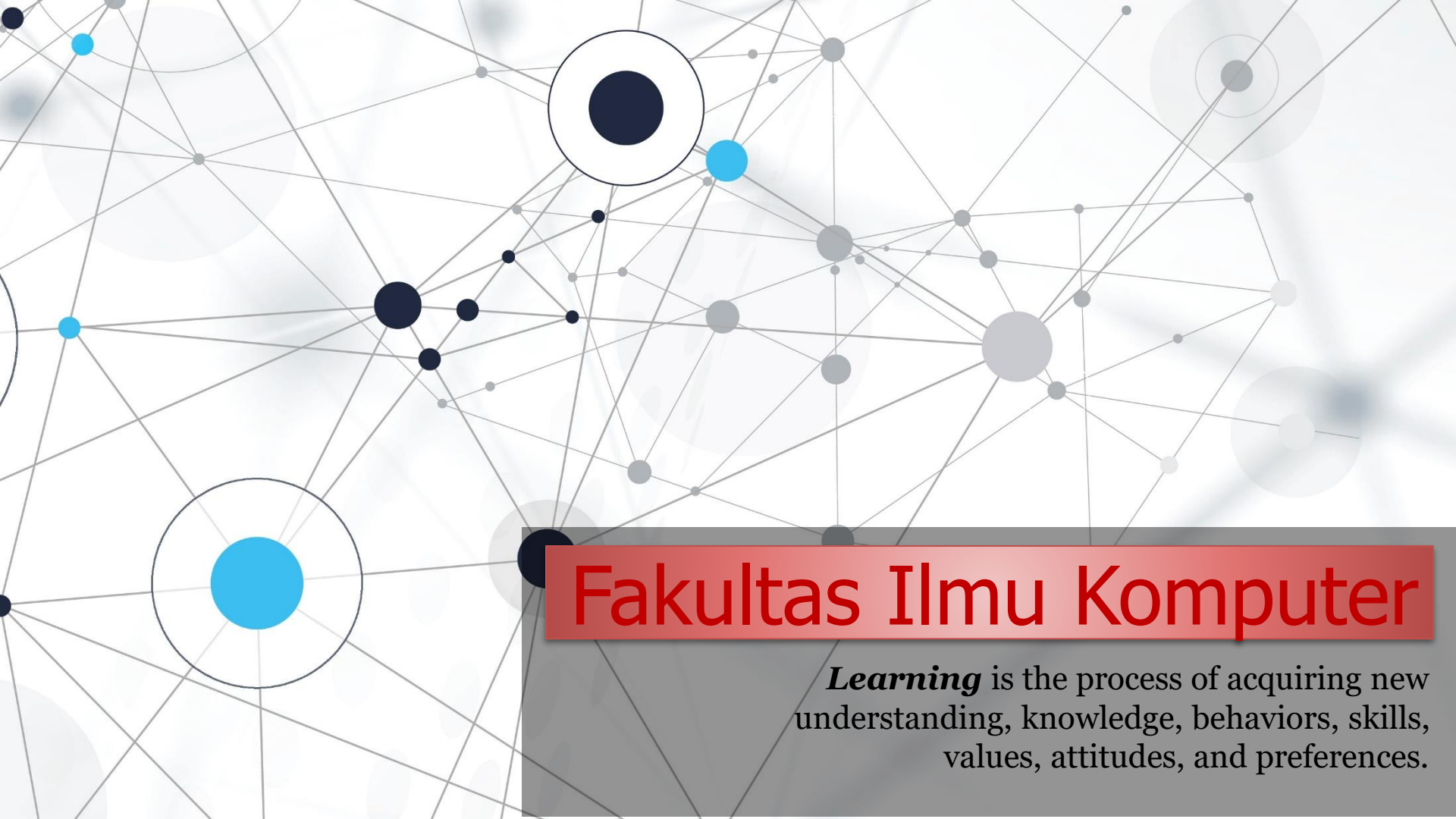


An abstract network diagram with various sized nodes (black, blue, and grey) connected by thin grey lines. Some nodes are highlighted with larger circles. The background is white with faint grey circles.

Fakultas Ilmu Komputer

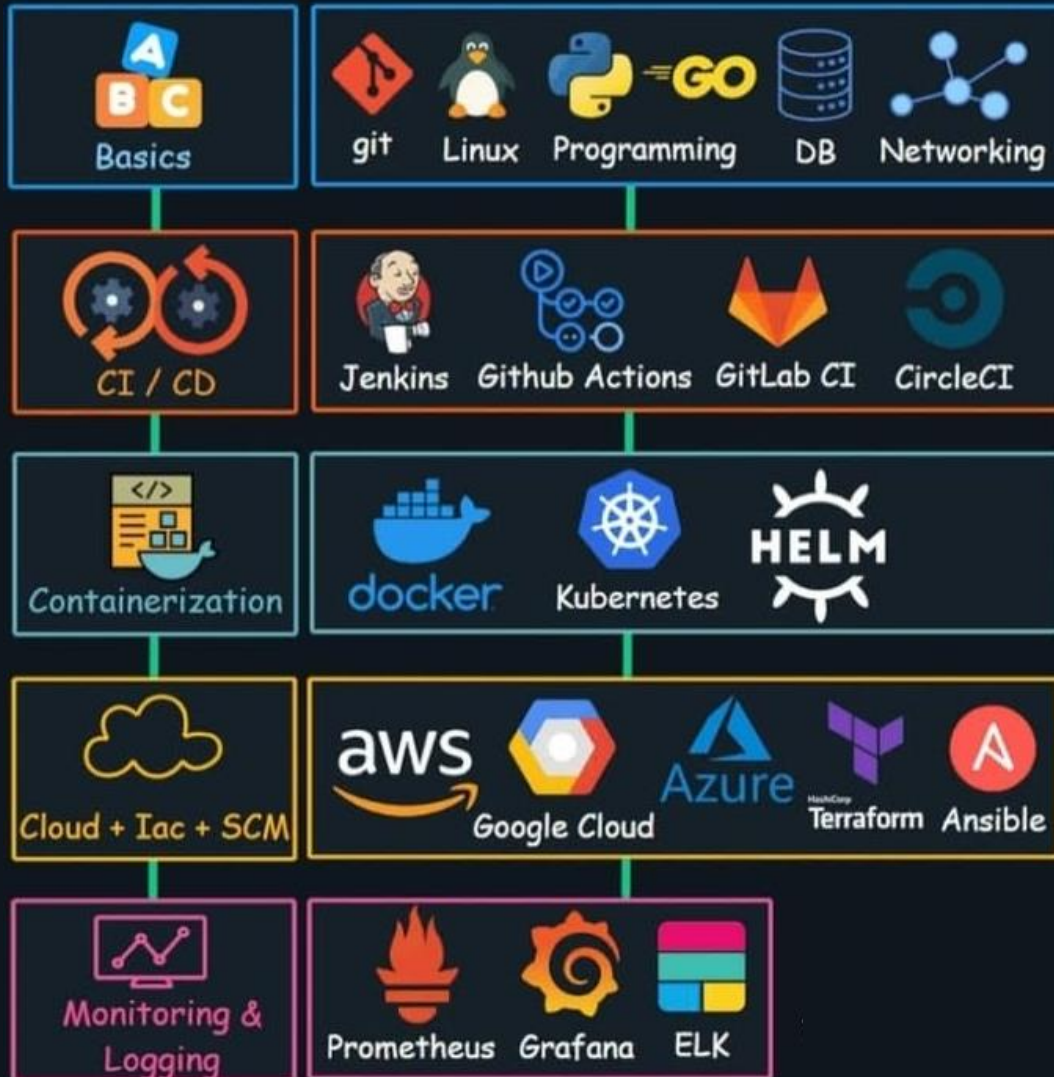
Learning is the process of acquiring new understanding, knowledge, behaviors, skills, values, attitudes, and preferences.

Microservices & Infrastructure as Code (IaC)



DevOps Learning Path

For **Beginners**



Typical SCM process:

(React, Vue, Svelte, atau Angular)

- 1) Developer **clones** a Git repository.
- 2) Makes changes to code.
- 3) **Commits** the changes locally with a message.
- 4) **Pushes** changes to a remote repository (e.g., GitHub).
- 5) A **Pull Request (PR)** or **Merge Request (MR)** is reviewed by teammates.
- 6) Once approved, changes are merged into the main branch.
- 7) The CI/CD pipeline (e.g., Jenkins, GitHub Actions) automatically builds and deploys the new version.

From **Devops** to **MLOPS**

MLOps = ML + DEV + OPS



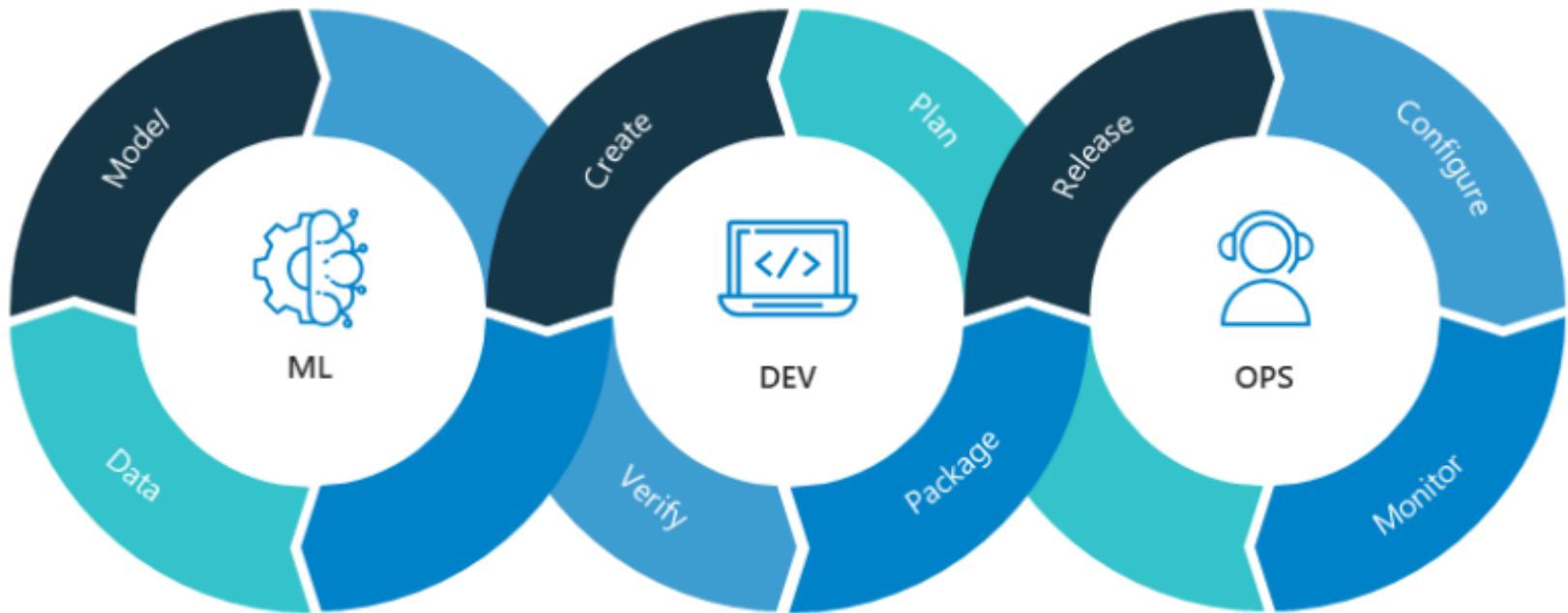
Develop

- Algorithm Training + Testing
- ETL (Data Pipelines)
- Continuous Integtation / Continuous Deployment

Operate

- Continuous Delivery
- Model Inference
- Monitoring and Management

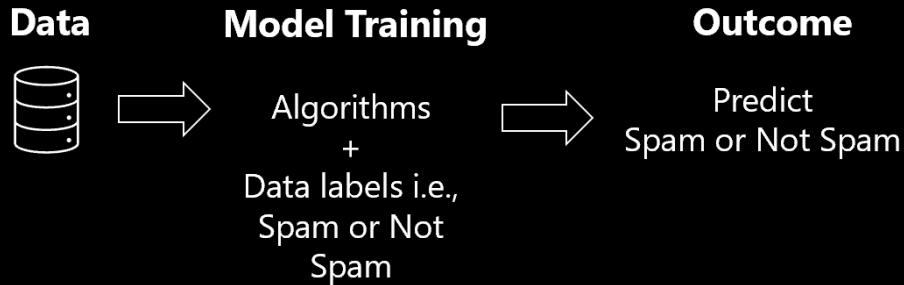
$$\text{MLOps} = \text{ML} + \text{DEV} + \text{OPS}$$



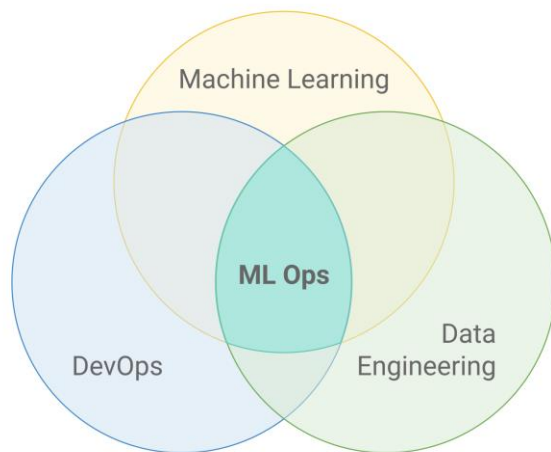
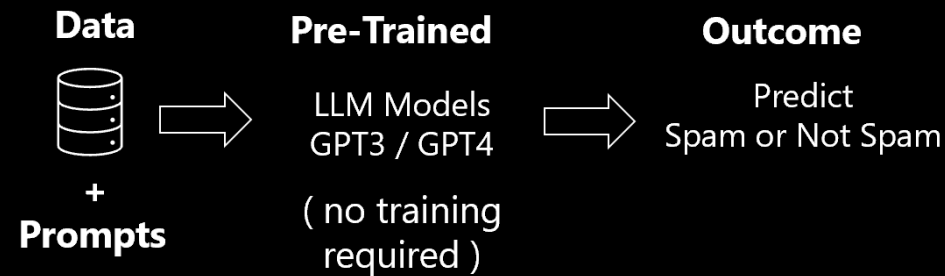
MLOps (Machine Learning Development Operations) adalah praktik (*practices*) dan metodologi (*methodologies*) yang digunakan untuk manage (*mengelola*), develop (*mengembangkan*), and deploy (*mendeploy*) solusi machine learning (ML) secara efisien dan berkelanjutan. MLOps menggabungkan konsep-konsep dari pengembangan sistem/software (DevOps) dengan tantangan khusus yang terkait dengan pengembangan model dan sistem ML.

Example: How to detect spam Emails?

ML Model



LLM Model



Large Language Model DevOps adalah pendekatan atau praktik yang mengintegrasikan large language model, seperti **GPT-3**, Google **PaLM 2**, Facebook **LLaMA 2** atau **GPT-4**, dalam lingkungan DevOps untuk mendukung pengembangan dan pengoperasian aplikasi yang melibatkan Natural Language Processing (*pemrosesan bahasa alami*) atau tugas terkait bahasa.

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1 – DevOps Microservices

(CI&CD Driven Microservice Deployments)

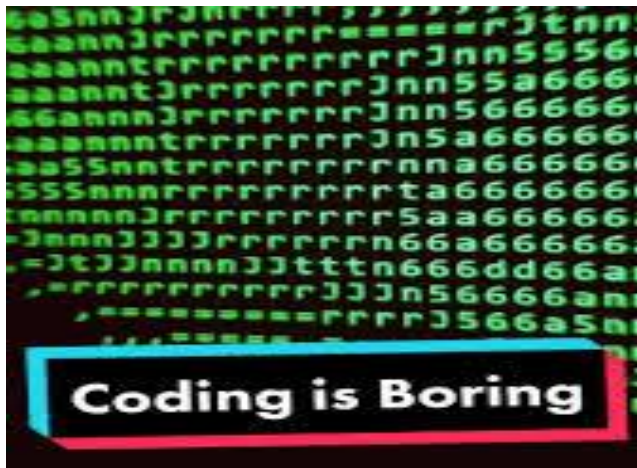
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Microservices

- ❖ “Creating a **system from a collection of small, isolated services, each of which owns their data, and is independently isolated** ... Services integrate with other services in order to form a cohesive system that’s far more flexible than the typical enterprise systems we build today.” [“Reactive Microservices Architecture” by Jonas Boner].
- ❖ The isolated service concept is what makes it easier to add features or fixes (*e.g. security fixes*) to a component of a system, without impacting other components of the same system (*where a “system” underlies a business process*).
- ❖ The opposite of a **microservices modeled system** is sometimes called a “**MONOLITH**”: a system made up of components that are tightly coupled with each other, usually reading/writing to a shared database.
- ❖ The “**tightly coupled**” aspect and the shared database is what makes monoliths hard to change (*e.g. getting all the component owners to agree on a work window in order to add a feature to one of the components, or, even more difficult: a schema change to that shared database*).
- ❖ But the current reality is that many large enterprises have a lot of monolith type systems.



What is Vibe Coding?



Andrej Karpathy 
@karpathy

There's a new kind of coding I call "vibe coding", where you fully give in to the vibes, embrace exponentials, and forget that the code even exists. It's possible because the LLMs (e.g. Cursor Composer w Sonnet) are getting too good. Also I just talk to Composer with SuperWhisper so I barely even touch the keyboard. I ask for the dumbest things like "decrease the padding on the

About

Andrej Karpathy is a Slovak-Canadian computer scientist who served as the director of artificial intelligence and Autopilot Vision at Tesla. He co-founded and formerly worked at OpenAI, where he specialized in deep learning and computer vision. [Wikipedia](#)

Born: October 23, 1986 (age 38 years), [Slovakia](#)

Education: [Stanford University](#) (2016) · [See more](#)

h-index: 11

Award: [Innovators Under 35](#) (2020)

Affiliation: Stanford University

Research interests: Machine Learning, Computer Vision, Artificial Intelligence

Doctoral advisor: [Fei-Fei Li](#)

Vibe coding is an [AI](#)-dependent [programming](#) technique where a person describes a problem in a few sentences as a [prompt](#) to a [large language model \(LLM\)](#) tuned for coding. The LLM generates [software](#), shifting the programmer's role from manual coding to guiding, testing, and refining the AI-generated [source code](#). **Wikipedia.**



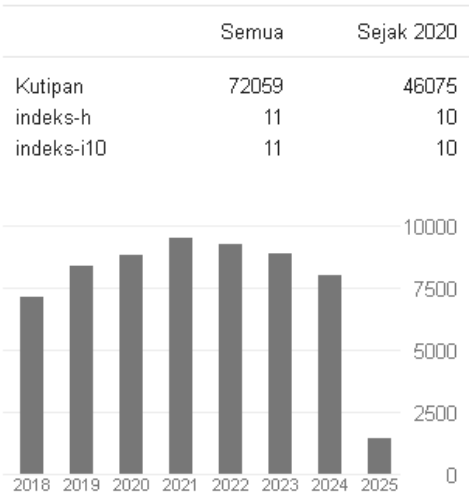
Andrej Karpathy

Computer Science PhD student, [Stanford University](#)
Email yang diverifikasi di [cs.stanford.edu](#) - [Beranda](#)
[Machine Learning](#) [Computer Vision](#) [Artificial Intelligence](#)



JUDUL	DIKUTIP OLEH	TAHUN
Imagenet large scale visual recognition challenge O Russakovsky, J Deng, H Su, J Krause, S Satheesh, S Ma, Z Huang, ... International Journal of Computer Vision 115 (3), 211-252	49792	2015
Large-scale video classification with convolutional neural networks A Karpathy, G Toderici, S Shetty, T Leung, R Sukthankar, L Fei-Fei Proceedings of the IEEE conference on Computer Vision and Pattern ...	8943	2014
Deep visual-semantic alignments for generating image descriptions A Karpathy, L Fei-Fei Proceedings of the IEEE Conference on Computer Vision and Pattern ...	7470	2015
Densecap: Fully convolutional localization networks for dense captioning J Johnson, A Karpathy, L Fei-Fei Proceedings of the IEEE Conference on Computer Vision and Pattern ...	1519	2016
Visualizing and Understanding Recurrent Networks A Karpathy, J Johnson, L Fei-Fei arXiv preprint arXiv:1506.02078	1505	2015
Deep fragment embeddings for bidirectional image sentence mapping A Karpathy, A Joulin, L Fei Fei Advances in Neural Information Processing Systems, 1889-1897	1130	2014
Grounded Compositional Semantics for Finding and Describing Images with Sentences R Socher, A Karpathy, Q Le, C Manning, A Ng Transactions of the Association for Computational Linguistics 2014	1065	2014
Locomotion skills for simulated quadrupeds S Coros, A Karpathy, B Jones, L Reveret, M Van De Panne ACM Transactions on Graphics (TOG) 30 (4), 59	250	2011
Object discovery in 3d scenes via shape analysis A Karpathy, S Miller, L Fei-Fei 2013 IEEE International Conference on Robotics and Automation, 2088-2095	226	2013
Curriculum learning for motor skills A Karpathy, M Van De Panne Canadian Conference on Artificial Intelligence, 325-330	81	2012

Dikutip oleh [LIHAT SEMUA](#)



Introduction of Microservices

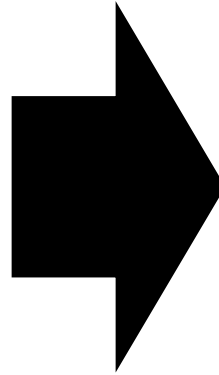
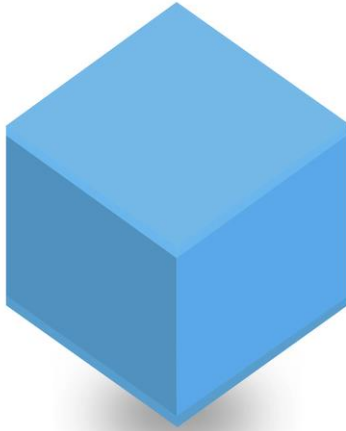
MICROSERVICES architecture is an architectural style that allows to develop an application as a compilation of multiple services instead of developing a single application (*monolith*).

The architectural style offers several advantages over **MONOLITH**, including:-

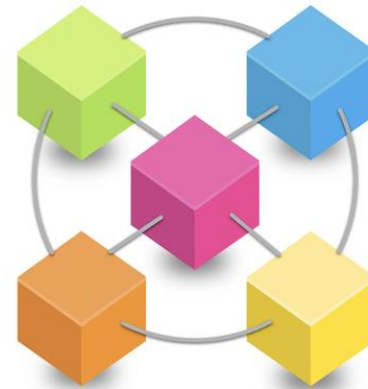
- ❖ Easy to test and maintain (*Mudah diuji dan dirawat*)
- ❖ Loosely joined (*Bergabung secara longgar*)
- ❖ Separately deployable (*Dapat diterapkan secara terpisah*)
- ❖ Planned around business effectiveness (*Direncanakan berdasarkan efektivitas bisnis*)
- ❖ Managed by a small team (*Dikelola oleh tim kecil*)

Monolith, Microservice, dan DevOps adalah konsep dan praktik yang berhubungan dengan pengembangan software (perangkat lunak).

Monolithic



Microservices

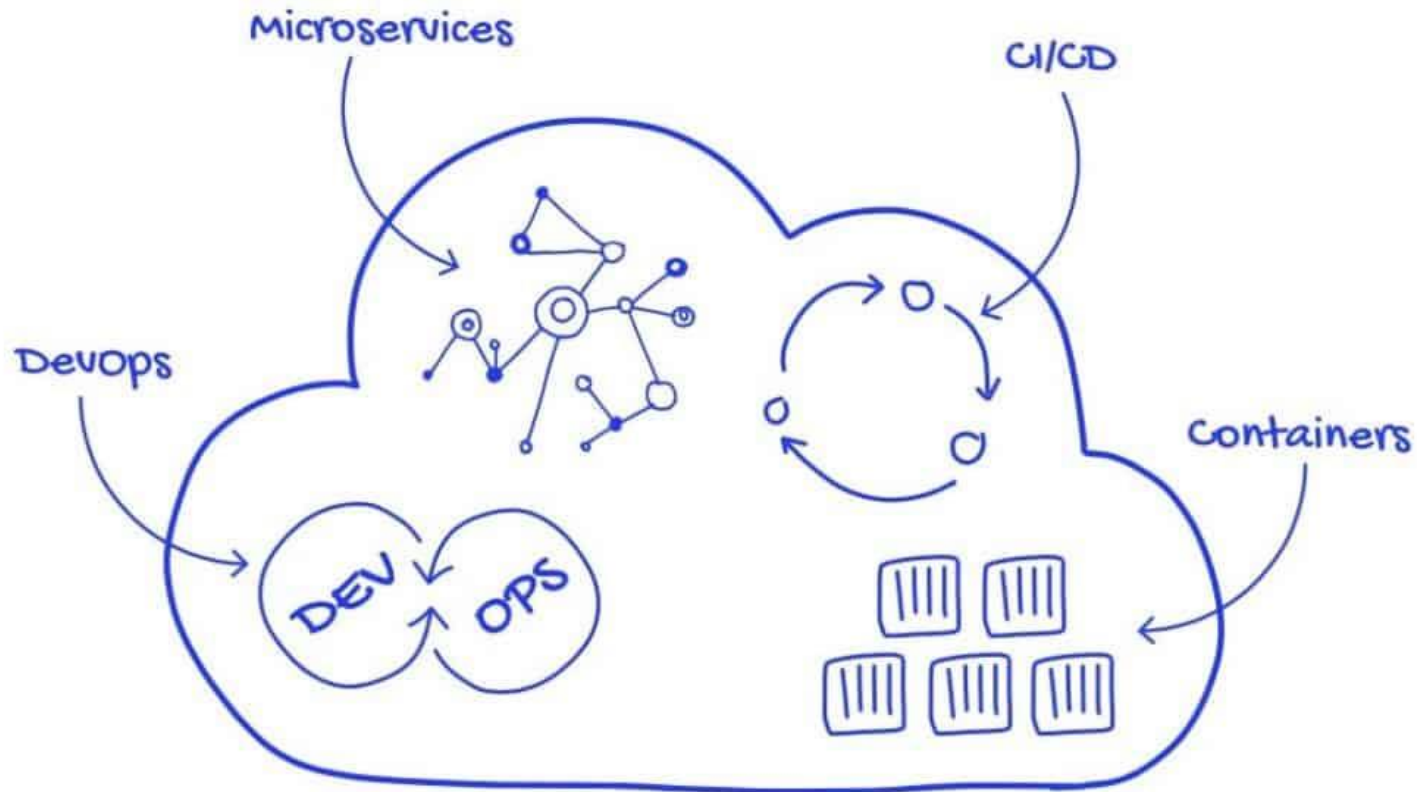


Monolith (Monolithic)

Monolithic is a software architecture pattern where the entire application is built as a single unit. This means all the components and features of the application are contained in a single source code and are usually worked on as one large project. Monolithic is generally easier to start, but becomes difficult to maintain and develop as the application grows. **A change in one part can affect the entire application.**

Microservice:

Microservices are a software architecture pattern in which an application is divided into various components called "microservices". Each microservice is an independent unit that has specific responsibilities, and they communicate through a network interface (usually HTTP). This approach allows for easier scalability, maintenance, and development, but it also requires managing the complexity of managing multiple services.



DevOps and **Microservices** are the foundation of modern software development practices and cloud-native applications. Microservices, along with DevOps, lay the foundation for designing agile software applications for enterprises. **DevOps and microservices architecture allows organizations to deploy cloud-based applications as independent services**. As microservices and DevOps are complementing each other, enterprises achieve greater agility and operational efficiency with these modern application development and deployment methodologies.

[1] DevOps



- 1) DevOps is a **collaborative approach** that combines development and operations in the software development lifecycle.
- 2) The goal is **to accelerate (mempercepat) software delivery**, improve quality and streamline work between developers and operations teams with automation and continuous integration.
- 3) DevOps focuses on **culture**, **processes** and **tools** to achieve faster and more reliable delivery.

[2] CI/CD

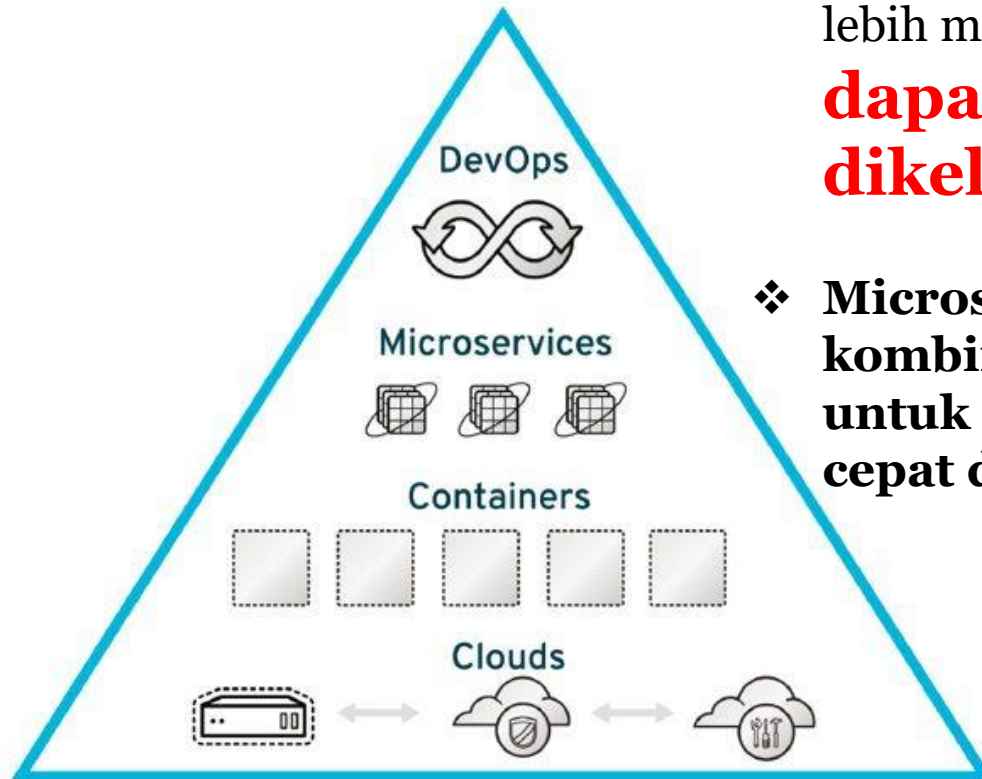
CI/CD (Continuous Integration dan Continuous Delivery):

- 1) CI adalah **automation practice** di mana perubahan kode disatukan secara teratur ke dalam shared repository (*repositori bersama*), tested (*diuji*) dan evaluated (*dievaluasi*) untuk mendeteksi masalah secepat mungkin.
- 2) CD adalah kelanjutan dari CI dan mencakup pengiriman (Continuous Delivery) dan pengiriman lanjutan (Continuous Deployment).
- 3) **Continuous Delivery** is the practice of automating the process of delivering to a test or production environment, while Continuous Deployment is the practice of automating delivery directly to production after passing automation tests.
(*Continuous Delivery adalah praktik mengotomatiskan proses pengiriman ke lingkungan pengujian atau produksi, sementara Continuous Deployment adalah praktik yang mengotomatiskan pengiriman langsung ke produksi setelah melewati tes otomatisasi*).

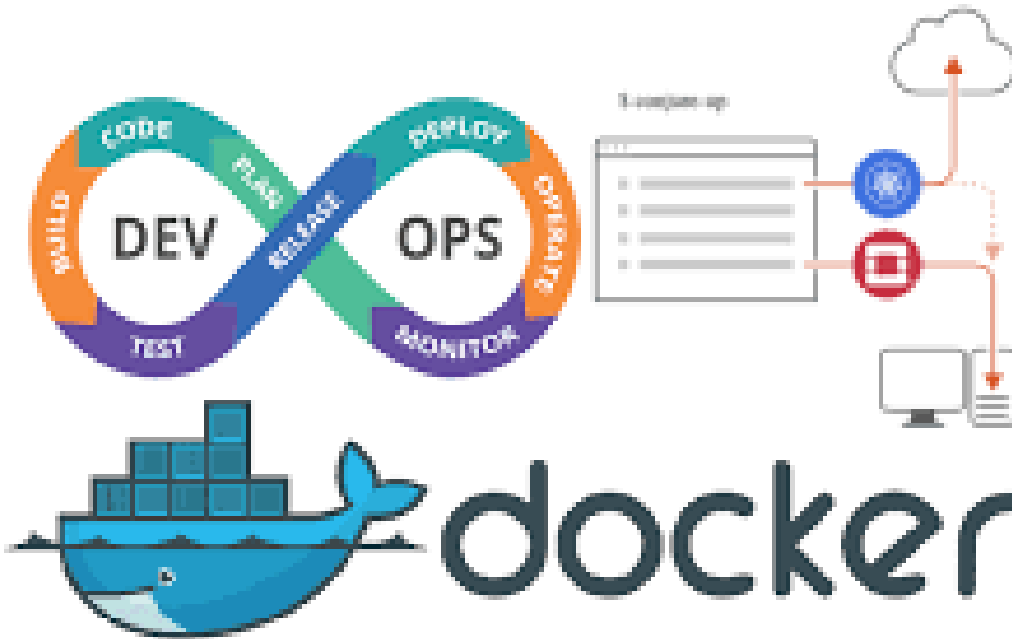
[3] Microservices

Microservices is a **software architecture approach** in which applications are built as a number of small, standalone services that communicate through open *interfaces* and can be managed independently.

- ❖ Ini memungkinkan **skalabilitas**, **pemeliharaan**, dan **pengembangan** yang lebih mudah karena **setiap layanan dapat dikembangkan dan dikelola secara terpisah**.
- ❖ Microservices sering digunakan dalam kombinasi dengan DevOps dan CI/CD untuk mencapai pengiriman yang lebih cepat dan lebih fleksibel.



[4] Containers



Containers are packaged units that contain software and all its dependencies, including code, libraries, and system settings. They allow applications to be well isolated and can run consistently across multiple environments, from development to production.

Teknologi seperti Docker umumnya digunakan untuk membuat dan mengelola **containers**. Containers sering digunakan bersamaan dengan Microservices untuk menjaga aplikasi tetap terisolasi dan mudah dikelola.

Monolithic

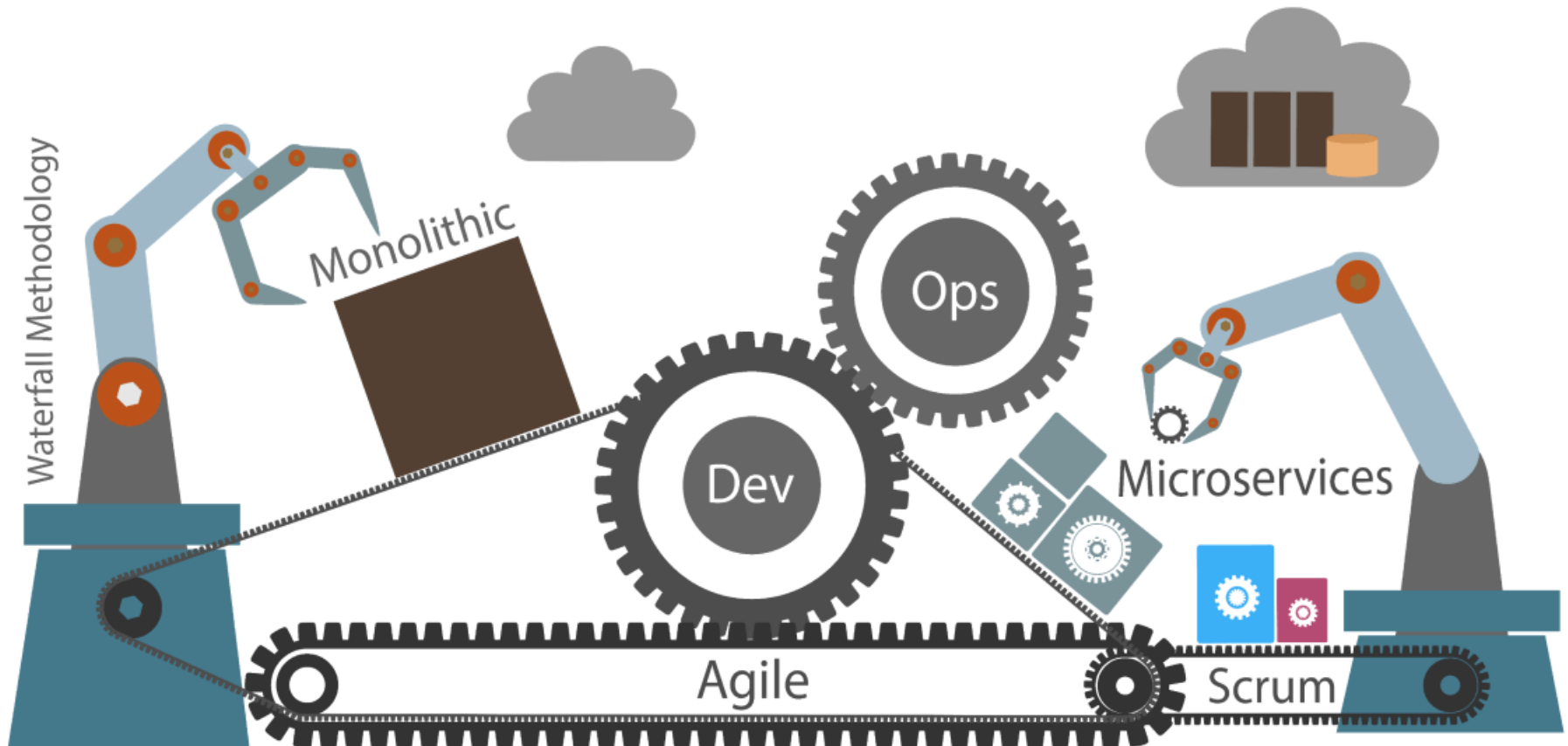
Microservices

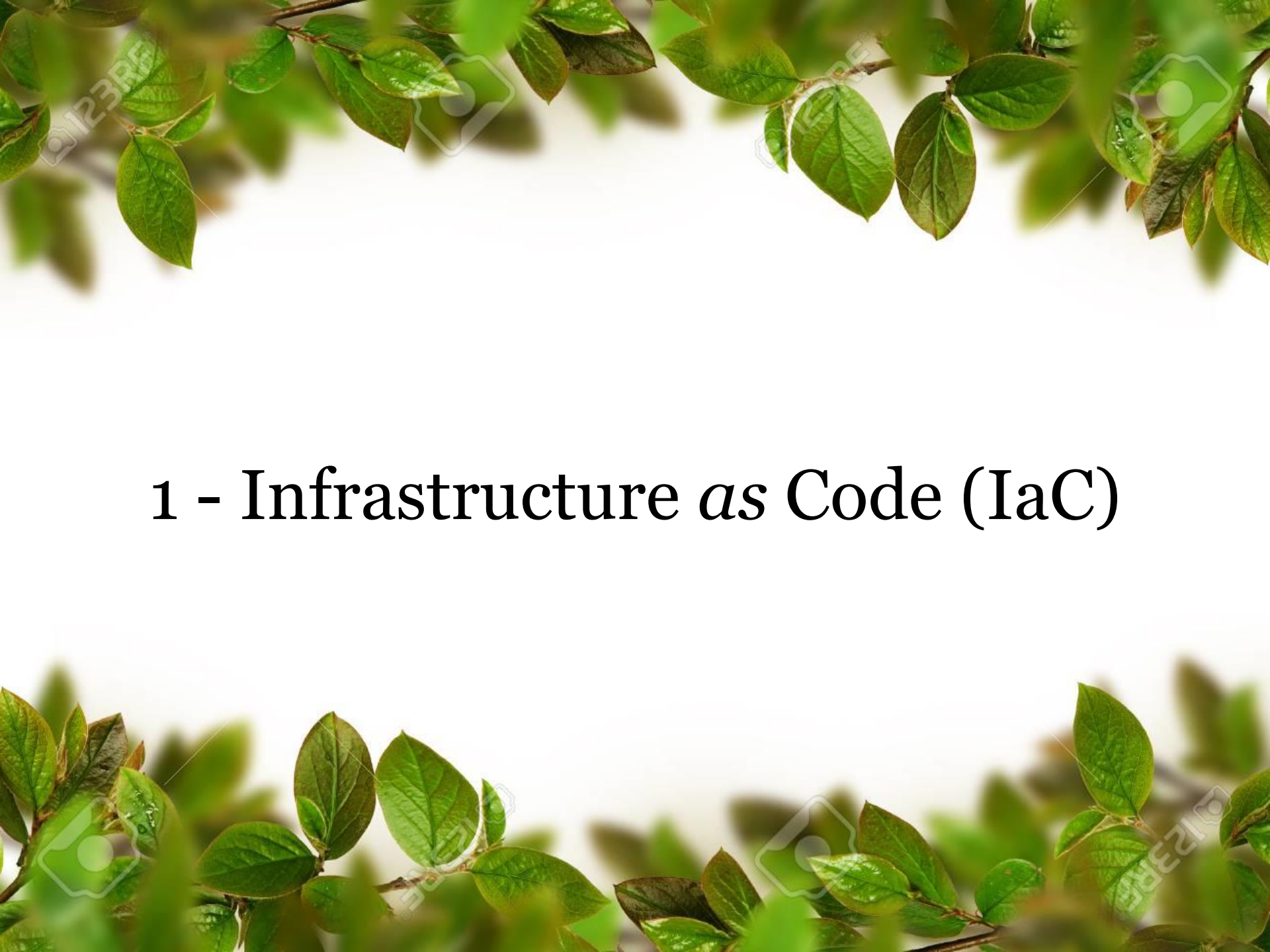
Serverless
architecture for software



When implemented correctly, DevOps, Microservices, Containers, and Cloud Computing allow an IT team to simultaneously **optimize** for all three of the traditional project constraints: **Cost, Quality, and Time to Market.**

- ❖ **Agile** adalah kerangka kerja pengembangan *software* yang berfokus pada *fleksibilitas*, *adaptabilitas*, dan *kolaborasi*. Agile mempromosikan prinsip-prinsip seperti mengutamakan individu dan interaksi, *software* yang berfungsi, kolaborasi pelanggan, serta merespons perubahan. Agile memungkinkan tim pengembangan untuk lebih mudah menyesuaikan diri dengan perubahan kebutuhan pelanggan dan market.
- ❖ **Scrum** adalah salah satu metode pengelolaan *project* yang digunakan dalam kerangka kerja Agile. Scrum memiliki aturan dan peran yang ketat, termasuk Scrum Master, Product Owner, dan tim pengembangan. Scrum **mengatur pekerjaan ke dalam periode waktu yang disebut sprint** (biasanya 2-4 minggu), di mana tim bekerja untuk menghasilkan inkrement produk yang dapat digunakan. Scrum mempromosikan *transparansi*, *inspeksi* dan *adaptasi*, yang merupakan prinsip-prinsip Agile.



The slide features a white background with green leaves and branches framing the top and bottom edges. A large, faint watermark with the text '123RF' is visible diagonally across the slide.

1 - Infrastructure *as* Code (IaC)

Learning DevOps

The complete guide to accelerate collaboration with Jenkins, Kubernetes, Terraform and Azure DevOps

Packt>

www.packt.com

Mikael Krief

1

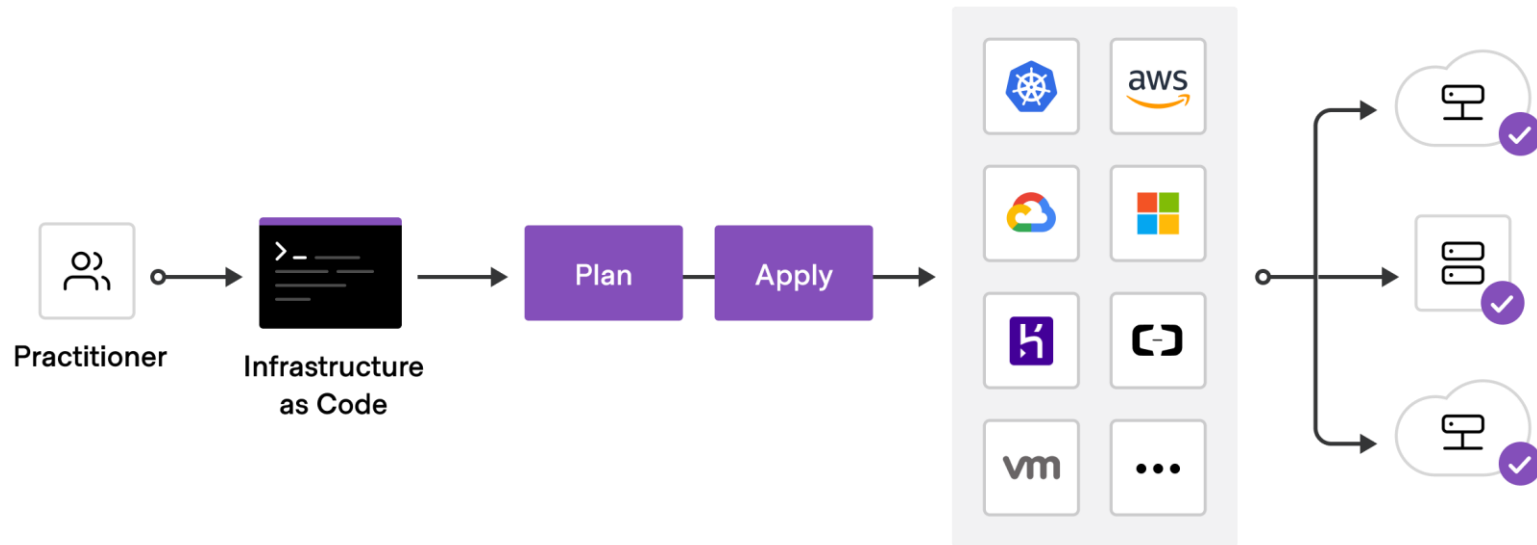
Section 1: DevOps and Infrastructure as Code

The objectives of the first section are to present the DevOps culture and to provide all of the keys for the best Infrastructure as Code practices. This section explains the DevOps application on cloud infrastructure, showing provisioning using Terraform and configuration with Ansible. Then, we improve this by templating this infrastructure with Packer.

We will have the following chapters in this section:

- Chapter 1, *DevOps Culture and Practices*
- Chapter 2, *Provisioning Cloud Infrastructure with Terraform*
- Chapter 3, *Using Ansible for Configuring IaaS Infrastructure*
- Chapter 4, *Optimizing Infrastructure Deployment with Packer*

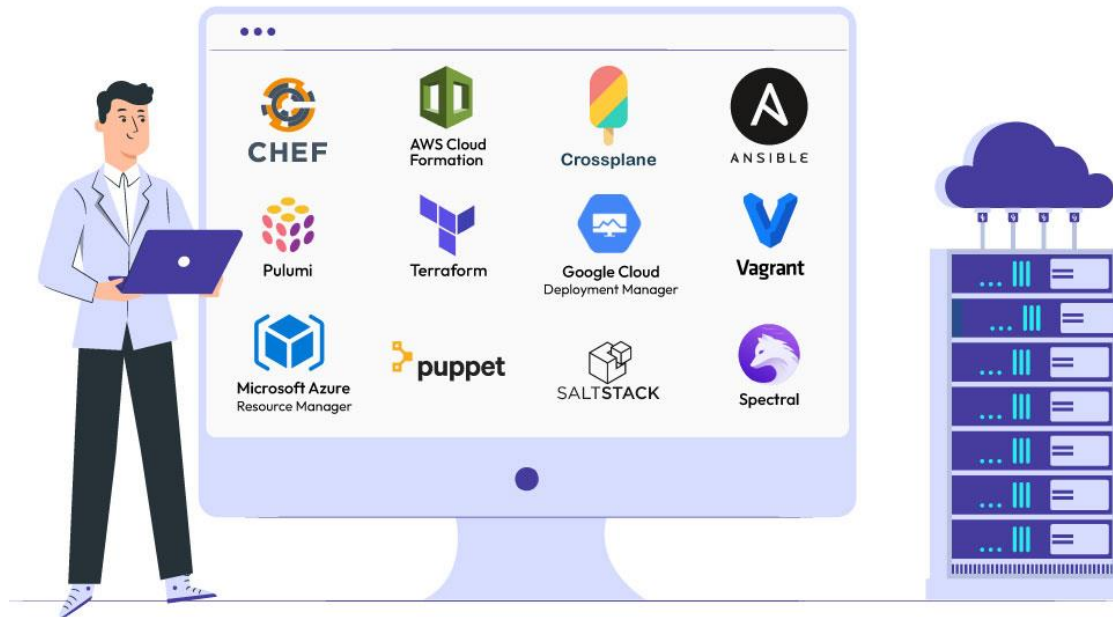
Infrastructure *as* Code (IaC)



DevOps is a culture different from traditional corporate cultures and requires a change in mindset, processes and tools. It is often associated with continuous integration (CI) and continuous delivery (CD) practices, which are software engineering practices, but also with Infrastructure as Code (IaC), which consists of **codifying the structure and configuration of infrastructure**.

Infrastructure as Code (IaC) memiliki arti bahwa infrastruktur, seperti server, networking, database, dan sumber daya lainnya, didefinisikan dan dikonfigurasi menggunakan **computer code** daripada secara manual.

Infrastructure *as* Code (IaC)



Penggunaan IaC membantu dalam menciptakan infrastruktur yang konsisten, mengurangi kesalahan manusia, mempercepat pengiriman aplikasi, serta memungkinkan pengujian dan pengelolaan infrastruktur yang lebih efisien.

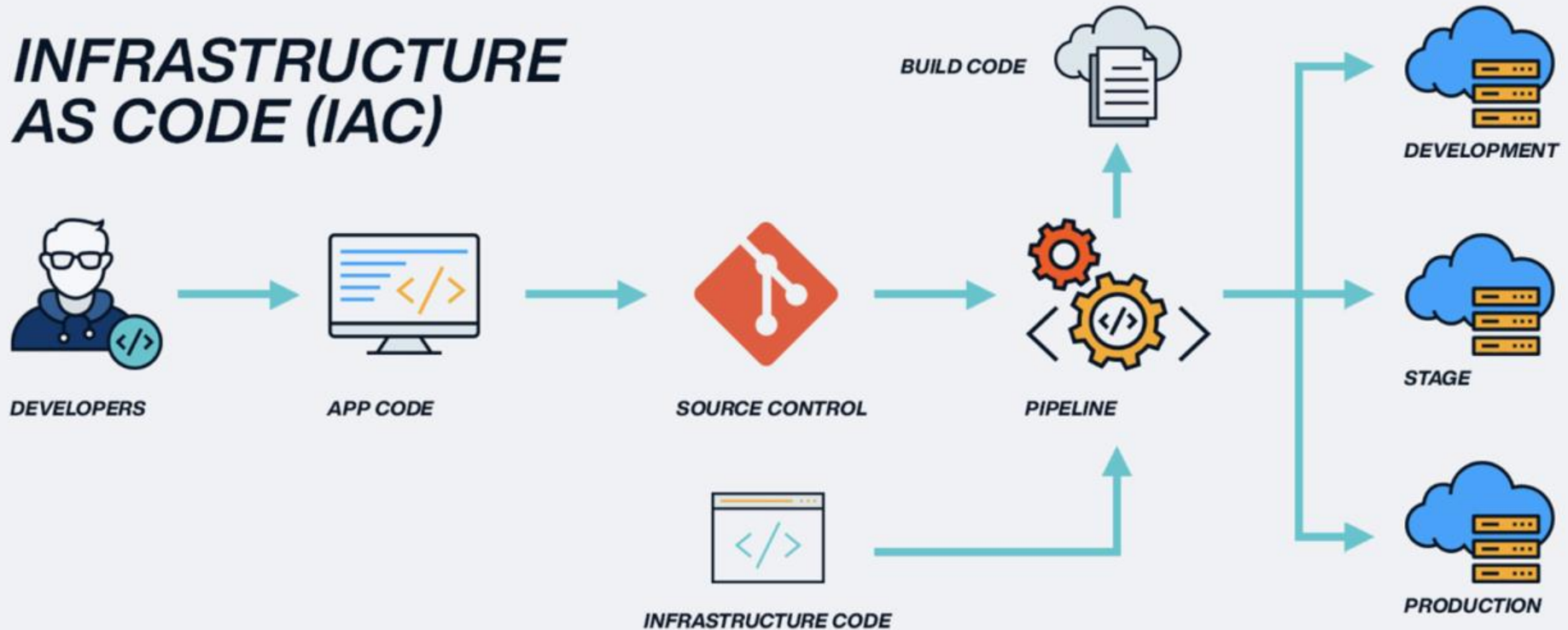
What is IaC?

Infrastructure as Code (IaC) adalah proses pengelolaan (*managing*) dan penyediaan (*provisioning*) infrastruktur menggunakan **machine-readable definition files** (*file definisi yang dapat dibaca oleh mesin*), bukan konfigurasi manual. Dengan IaC, sebuah organisasi dapat mendefinisikan infrastruktur dalam file teks, yang bisa dikontrol versinya dan digunakan secara otomatis menggunakan tools seperti Terraform, Ansible dan Chef.

IaC provides several benefits, including:

1	Version Control	Infrastructure definitions can be version controlled, which <u>provides traceability and accountability for infrastructure changes</u> . Version control systems enable us to see who made the changes, when they made them, and what changes were made.
2	Reusability	IaC enables organizations <u>to reuse code for infrastructure components like servers, databases, and networks</u> . This helps to <u>reduce duplication</u> of effort and increases efficiency.
3	Consistency	Infrastructure definitions can be tested and validated before deployment, which ensures consistency in the environment. This <u>helps to prevent issues caused by manual configuration errors</u> .

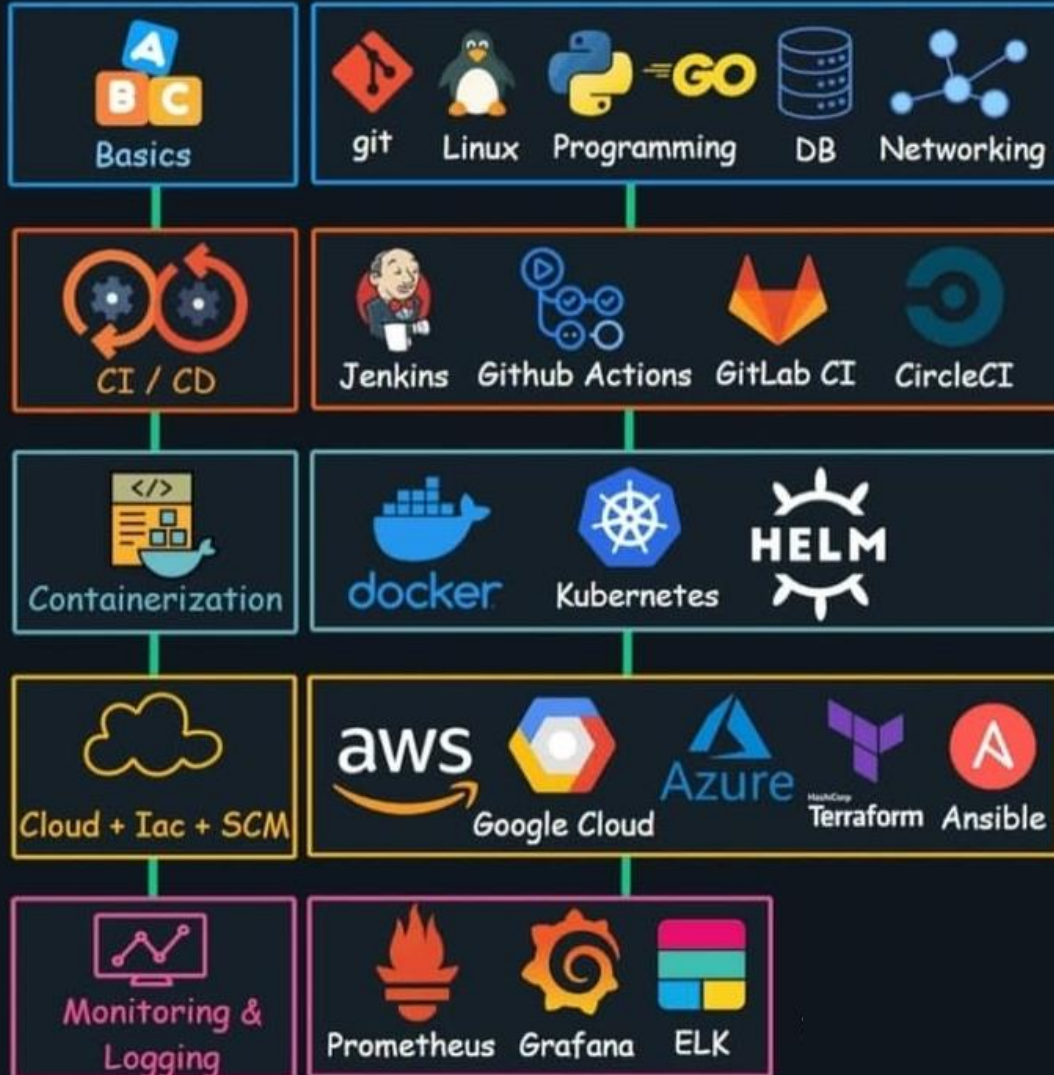
INFRASTRUCTURE AS CODE (IAC)



Dengan Infrastructure as Code (IaC), tim DevOps dapat mendefinisikan infrastruktur yang diperlukan untuk aplikasi dalam kode, yang kemudian dapat disimpan dalam repositori kode sumber dan dikelola melalui siklus pengembangan software, termasuk develop (*pembangunan*), testing (*pengujian*), dan delivery (*pengiriman*). Infrastructure as Code (IaC) dapat **membantu mempercepat proses pengembangan dan meminimalkan masalah yang mungkin timbul akibat perbedaan** antara lingkungan Development (*pengembangan*), testing (*pengujian*), and production (*produksi*).

DevOps Learning Path

For **Beginners**



Typical SCM process:

- 1) Developer **clones** a Git repository.
- 2) Makes changes to code.
- 3) **Commits** the changes locally with a message.
- 4) **Pushes** changes to a remote repository (e.g., GitHub).
- 5) A **Pull Request (PR)** or **Merge Request (MR)** is reviewed by teammates.
- 6) Once approved, changes are merged into the main branch.
- 7) The CI/CD pipeline (e.g., Jenkins, GitHub Actions) automatically builds and deploys the new version.

How to Enable IaC and CI/CD?

To enable IaC and CI/CD, organizations need to follow these steps:

Step 1	Define Infrastructure as Code (IaC)	✓ The <i><u>first step</u></i> is to define infrastructure as code using tools like Terraform, Ansible, or Chef. ✓ With IaC, infrastructure can be defined in a text file, which can be version controlled and deployed automatically.
Step 2	Create a CI/CD Pipeline	✓ The <i><u>second step</u></i> is to create a CI/CD pipeline using tools like Jenkins, GitLab CI/CD, or Azure DevOps. ✓ With a CI/CD pipeline, code changes can be automatically tested, built, and deployed to production.
Step 3	Integrate IaC and CI/CD	✓ The <i><u>final step</u></i> is to integrate IaC and CI/CD. ✓ This can be done by deploying infrastructure changes automatically as part of the CI/CD pipeline. ✓ This ensures that the infrastructure is consistent and up-to-date with the latest code changes.



Exercise *for* Students

[1] Start Jenkins

Saya install di *root directory*, jadi saat *restart* harus masuk ke *directory* yang sama. Selanjutnya type:

- **sudo su**
- **sudo java -jar jenkins.war --httpPort=9090 &**

```
root@devops:/home/semmy# ls
root@devops:/home/semmy# ll
total 32
drwxr-x--- 4 semmy semmy 4096 Sep 18 10:04 ./
drwxr-xr-x 3 root  root  4096 Sep 18 09:16 ../
-rw----- 1 semmy semmy   8 Sep 18 10:04 .bash_history
-rw-r--r-- 1 semmy semmy 220 Jan  6  2022 .bash_logout
-rw-r--r-- 1 semmy semmy 3771 Jan  6  2022 .bashrc
drwx----- 2 semmy semmy 4096 Sep 18 09:18 .cache/
-rw-r--r-- 1 semmy semmy 807 Jan  6  2022 .profile
drwx----- 2 semmy semmy 4096 Sep 18 09:16 .ssh/
root@devops:/home/semmy# cd ..
root@devops:/home# cd ..
root@devops:/# ls
bin  etc  lib  libx32  mnt  root  snap  tmp
boot nome lib32  lost+found  opt  run  srv  usr
dev  jenkins.war lib64  media  proc  sbin  sys  var
root@devops:/# cd home/
root@devops:/home# cd root
bash: cd: root: No such file or directory
root@devops:/home# cd ..
root@devops:/# sudo java -jar jenkins.war --httpPort=9090 &
```

Akan Muncul Error jika directory tidak sesuai:

- *Error: Unable to access jarfile jenkins.war*

```
root@devops:/home/semmy# ls
root@devops:/home/semmy# ll
total 32
drwxr-x--- 4 semmy semmy 4096 Sep 18 10:04 ./
drwxr-xr-x 3 root  root  4096 Sep 18  09:16 ../
-rw----- 1 semmy semmy    8 Sep 18 10:04 .bash_history
-rw-r--r-- 1 semmy semmy  220 Jan  6  2022 .bash_logout
-rw-r--r-- 1 semmy semmy 3771 Jan  6  2022 .bashrc
drwx----- 2 semmy semmy 4096 Sep 18  09:18 .cache/
-rw-r--r-- 1 semmy semmy  807 Jan  6  2022 .profile
drwx----- 2 semmy semmy 4096 Sep 18  09:16 .ssh/
root@devops:/home/semmy# cd ..
root@devops:/home# cd ..
root@devops:/# ls
bin  etc  lib  libx32  mnt  root  snap  tmp
boot  nome  lib32  lost+found  opt  run  srv  usr
dev  jenkins.war  lib64  media  proc  sbin  sys  var
root@devops:/# cd home/
root@devops:/home# cd root
bash: cd: root: No such file or directory
root@devops:/home# cd ..
root@devops:/# sudo java -jar jenkins.war --httpPort=9090 &
2023-09-24 11:57:44.417+0000 [id=23] INFO winstone.L
2023-09-24 11:57:45.115+0000 [id=30] INFO jenkins.Ini
2023-09-24 11:57:45.136+0000 [id=31] INFO jenkins.Ini
2023-09-24 11:57:47.140+0000 [id=30] INFO jenkins.Ini
2023-09-24 11:57:47.149+0000 [id=29] INFO jenkins.Ini
2023-09-24 11:57:47.184+0000 [id=31] INFO jenkins.Ini
2023-09-24 11:57:47.888+0000 [id=31] INFO jenkins.Ini
2023-09-24 11:57:47.889+0000 [id=28] INFO jenkins.Ini
2023-09-24 11:57:47.890+0000 [id=29] INFO jenkins.Ini
2023-09-24 11:57:47.892+0000 [id=29] INFO jenkins.Ini
2023-09-24 11:57:48.074+0000 [id=44] INFO hudson.util.Retrier#start: Attempt #1 to do the action check updates server
WARNING: An illegal reflective access operation has occurred
WARNING: Illegal reflective access by org.codehaus.groovy.vmplugin.v7.Java7$1 (file:/root/.jenkins/war/WEB-INF/lib/groovy-all-2.4.21.jar) to constructor java.lang.invoke.MethodHandles$Lookup(java.lang.Class,int)
WARNING: Please consider reporting this to the maintainers of org.codehaus.groovy.vmplugin.v7.Java7$1
WARNING: Use --illegal-access=warn to enable warnings of further illegal reflective access operations
WARNING: All illegal access operations will be denied in a future release
2023-09-24 11:57:48.711+0000 [id=28] INFO jenkins.InitReactorRunner$1#onAttained: Completed initialization
root@devops:/#
root@devops:/#
```

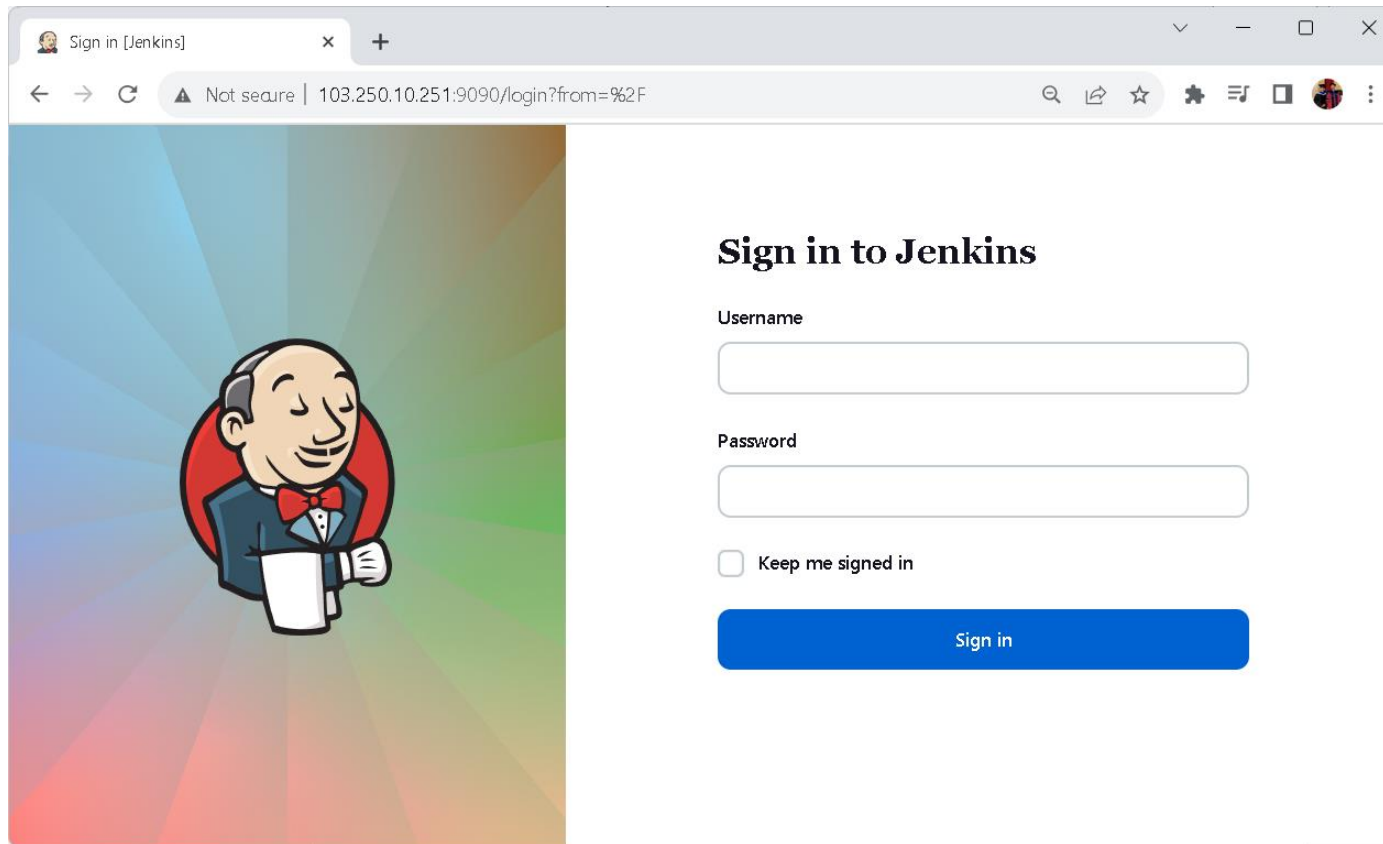
Jenkins Completed Initialization

```
root@devops: /  
s  
2023-09-24 11:57:44.417+0000 [id=23] INFO winstone.Logger#logInternal: Winstone Servlet Engine running: controlPort=disabled  
2023-09-24 11:57:45.115+0000 [id=30] INFO jenkins.InitReactorRunner$1#onAttained: Started initialization  
2023-09-24 11:57:45.136+0000 [id=31] INFO jenkins.InitReactorRunner$1#onAttained: Listed all plugins  
2023-09-24 11:57:47.140+0000 [id=30] INFO jenkins.InitReactorRunner$1#onAttained: Prepared all plugins  
2023-09-24 11:57:47.149+0000 [id=29] INFO jenkins.InitReactorRunner$1#onAttained: Started all plugins  
2023-09-24 11:57:47.184+0000 [id=31] INFO jenkins.InitReactorRunner$1#onAttained: Augmented all extensions  
2023-09-24 11:57:47.888+0000 [id=31] INFO jenkins.InitReactorRunner$1#onAttained: System config loaded  
2023-09-24 11:57:47.889+0000 [id=28] INFO jenkins.InitReactorRunner$1#onAttained: System config adapted  
2023-09-24 11:57:47.890+0000 [id=29] INFO jenkins.InitReactorRunner$1#onAttained: Loaded all jobs  
2023-09-24 11:57:47.892+0000 [id=29] INFO jenkins.InitReactorRunner$1#onAttained: Configuration for all jobs updated  
2023-09-24 11:57:48.074+0000 [id=44] INFO hudson.util.Retrier#start: Attempt #1 to do the action check updates server  
WARNING: An illegal reflective access operation has occurred  
WARNING: Illegal reflective access by org.codehaus.groovy.vmplugin.v7.Java7$1 (file:/root/.jenkins/war/WEB-INF/lib/groovy-all-2.4.21.jar) to con  
structor java.lang.invoke.MethodHandles$Lookup(java.lang.Class,int)  
WARNING: Please consider reporting this to the maintainers of org.codehaus.groovy.vmplugin.v7.Java7$1  
WARNING: Use --illegal-access=warn to enable warnings of further illegal reflective access operations  
WARNING: All illegal access operations will be denied in a future release  
2023-09-24 11:57:48.711+0000 [id=28] INFO jenkins.InitReactorRunner$1#onAttained: Completed initialization  
root@devops:/#  
root@devops:/# |
```

[2] Sign *Into* Jenkins 2023

Jenkins Default URL:

<http://103.250.10.251:9090/>



The screenshot shows a web browser window with the title "Sign in [Jenkins]". The address bar displays "Not secure | 103.250.10.251:9090/login?from=%2F". The page features a large illustration of the Jenkins mascot, a blue-suited man with a red bow tie, holding a white cup. To the right of the illustration, the heading "Sign in to Jenkins" is displayed. Below this heading are two input fields: "Username" and "Password". Under the "Password" field is a checkbox labeled "Keep me signed in". At the bottom of the form is a blue button labeled "Sign in".

Sign in [Jenkins]

Not secure | 103.250.10.251:9090/login?from=%2F

Sign in to Jenkins

Username

Password

☐ Keep me signed in

Sign in

Dashboard [Jenkins]

← → ↻ ⚠ Not secure | 103.250.10

Jenkins

Dashboard >

+ New Item

👤 People

📅 Build History

⚙ Manage Jenkins

📄 My Views

Build Queue

No builds in the queue.

Build Executor Status

1 Idle

2 Idle

103.250.10.251:9090/newJob

Sign in [Jenkins]

← → ↻ ⚠ Not secure | 103.250.10.251:9090/login?from=%2F



Sign in to Jenkins

Username

Password

☐ Keep me signed in

Sign in

Create a job →

Set up a distributed build

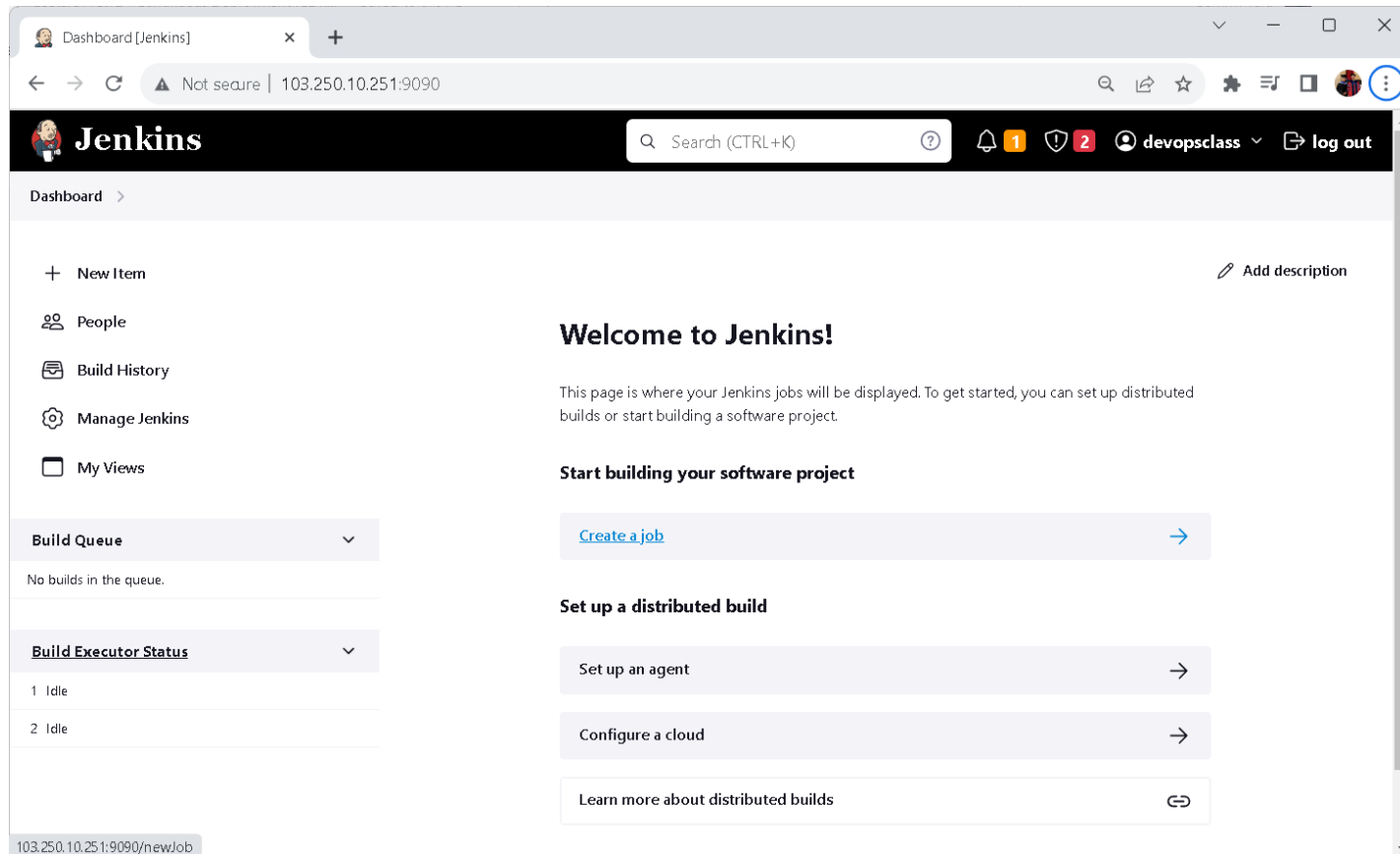
Set up an agent →

Configure a cloud →

Learn more about distributed builds ↗

[3] Dashboard Jenkins 2.414.1

Jenkins adalah salah satu alat yang sangat populer dalam ekosistem DevOps yang digunakan untuk otomatisasi berbagai tahap dari siklus pengembangan perangkat lunak, termasuk integrasi terus-menerus (Continuous Integration, CI) dan pengiriman terus-menerus (Continuous Delivery, CD). Jenkins memiliki antarmuka pengguna berbasis web yang disebut "Jenkins Dashboard" atau "Jenkins UI".



[1]

Install Git Plugin *in* Jenkins



Jenkins

Install Git Plugin

- ❖ Secara default, Jenkins tidak tahu bagaimana cara mengambil kode program yang terdapat di Git.
- ❖ Kita perlu menginstall Git Plugin terlebih dahulu.
- ❖ Setelah menginstall Git Plugin, maka secara otomatis kita bisa menambahkan sumber kode dari git repository di Job yang sudah kita buat.

The image shows two overlapping screenshots from the Jenkins web interface. The top screenshot is the 'System Configuration' page, which has a sidebar with icons for System, Plugins, Tools, Nodes, and Clouds. The 'Plugins' option is highlighted with a red rectangle. The bottom screenshot is the 'Available plugins' page, showing a search for 'git' and a list of results.

System Configuration

- System**
Configure global settings and paths.
- Plugins**
Add, remove, disable or enable plugins that can extend the functionality of Jenkins.
- Tools**
Configure tools, their locations and automatic installers.
- Nodes**
Add, remove, control and monitor the various nodes that Jenkins runs jobs on.
- Clouds**
Add, remove, and configure cloud instances to provision agents on-demand.

Available plugins - Plugins [Jenk x +]

Not secure | 103.250.10.251:9090/manage/pluginManager/ava

Jenkins

Search (CTRL+)

Dashboard > Manage Jenkins > Plugins

Plugins

Search: git

Install Name Released

☒	Git 5.2.0 git Source Code Management This plugin integrates Git with Jenkins.	3 mo 13 days ago
☐	Git client 4.5.0 Library plugins (for use by other plugins) Utility plugin for Git support in Jenkins	1 mo 9 days ago

Available plugins - Plugins [Jen] x +

← → ↻ ⚠ Not secure | 103.250.10.251:9

Jenkins

Dashboard > Manage Jenkins > Plugins

Updates

Available plugins

Installed plugins

Advanced settings

Plugins

git

Install

Install	Name	Released
☒	Git 5.2.0 git Source Code Management This plugin integrates Git with Jenkins.	3 mo 13 days ago
☐	Git client 4.5.0 Library plugins (for use by other plugins) Utility plugin for Git support in Jenkins	1 mo 9 days ago

System Configuration



System

Configure global settings and paths.



Tools

Configure tools, their locations and automatic installers.



Plugins

Add, remove, disable or enable plugins that can extend the functionality of Jenkins.



Nodes

Add, remove, control and monitor the various nodes that Jenkins runs jobs on.



Clouds

Add, remove, and configure cloud instances to provision agents on-demand.



Updates

Available plugins

Installed plugins

Advanced settings

Plugins

git

Install

Install after restart

Install	Name	Released
☒	Git 5.2.0 git Source Code Management This plugin integrates Git with Jenkins.	3 mo 13 days ago

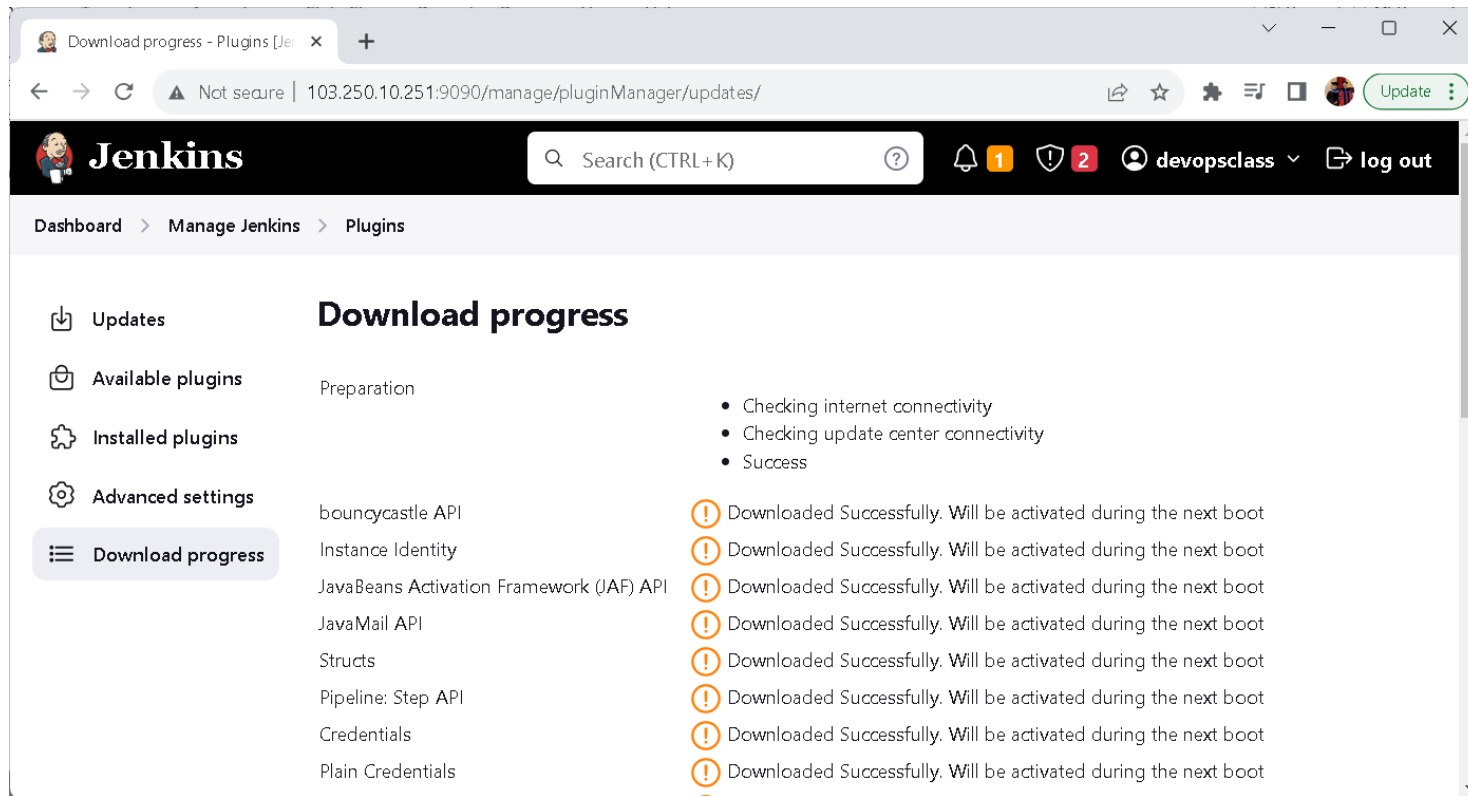
Install Git Plugin



Buka click **Dashboard** > "**Manage Jenkins**" > "**Plugins**".

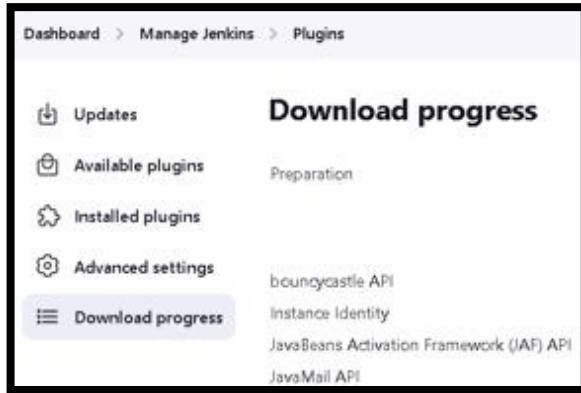
- 1) Di halaman "Manage Jenkins" dan pada bagian "**System Configuration**" pilih "**Plugins**" atau "Manage Plugins and Integrations".
- 2) Di jendela "Plugin Manager", pilih tab "**Available Plugins**". Ini akan menampilkan daftar plugin yang tersedia untuk diinstal.
- 3) Di kolom "Filter" atau "**Search**" di bagian atas, ketik "**Git**". Jenkins akan menyaring daftar plugin yang sesuai.
- 4) Temukan plugin "Git" dalam daftar hasil pencarian. **Centang kotak** di sebelah nama plugin dan click button "**Install After Restart**" di samping kiri.

Plugins Download Progress

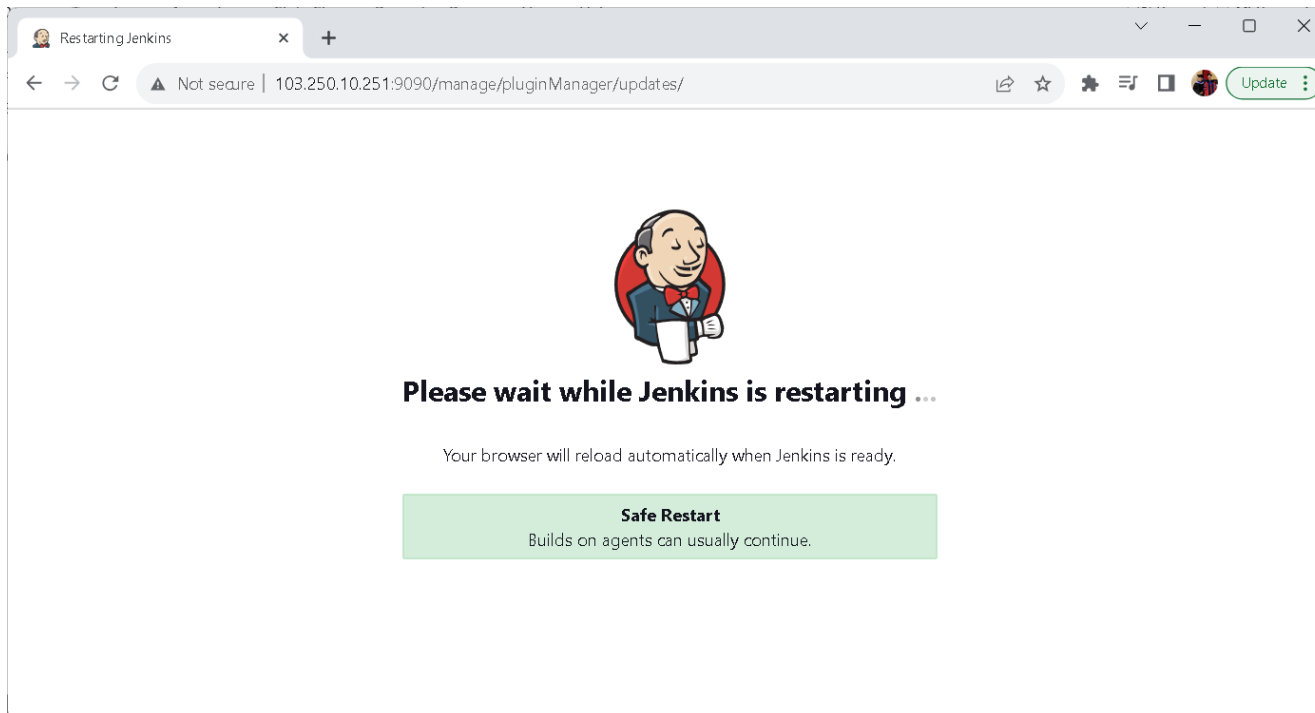


Fitur ini digunakan untuk mengecek status dan kemajuan download plugin dalam Jenkins. Ini adalah alat bantu yang memungkinkan kita untuk memantau dan mengevaluasi kemajuan saat Jenkins mengunduh dan menginstal plugin-plugin yang diperlukan.

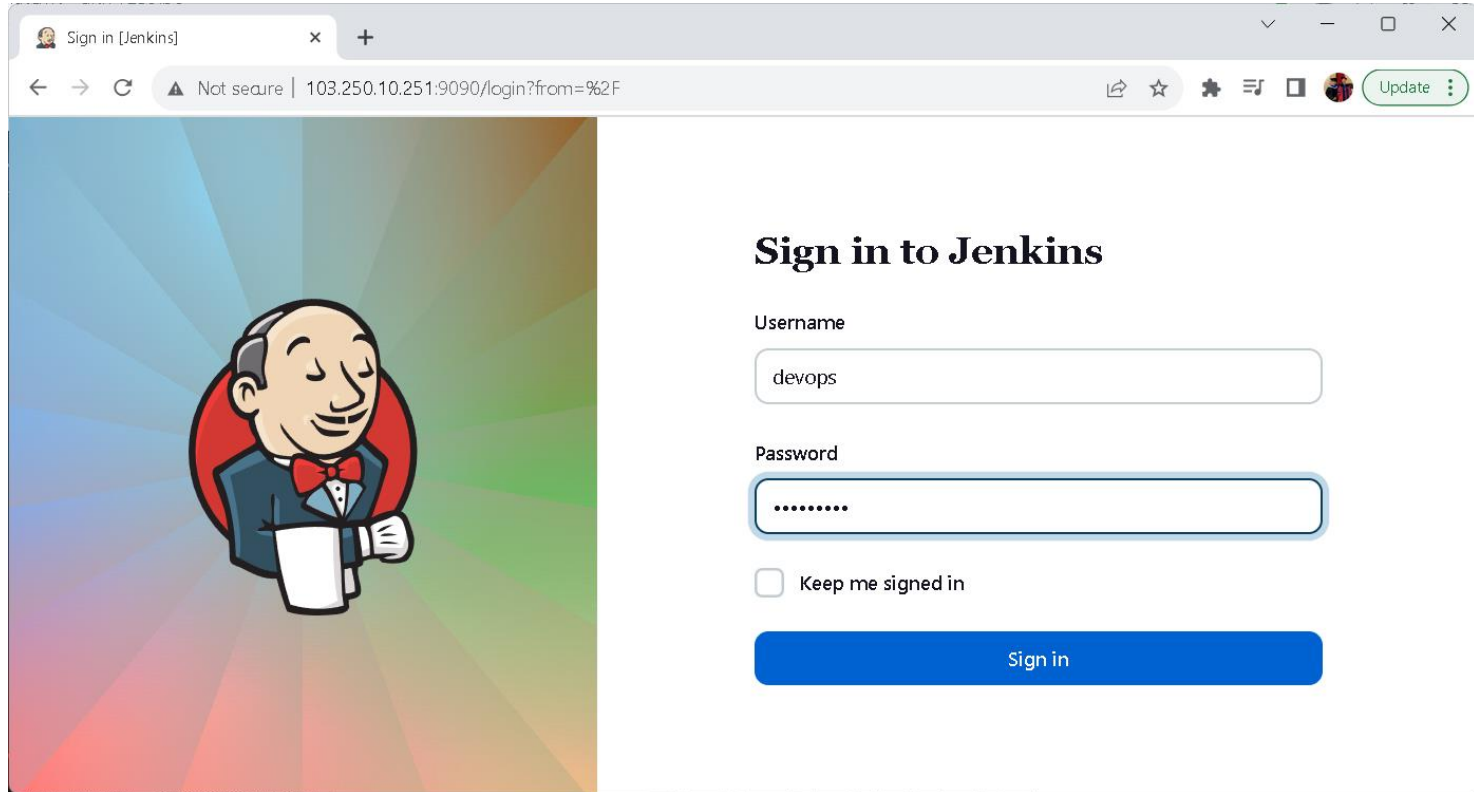
Restart Jenkins After Installation



Scroll ke bagian bawah gambar di samping dan pilih Check Box “**Restart Jenkins installation is complete and no jobs are running**”. Secara otomatis Jenkins akan restart.



After Restart, Login Jenkins



A screenshot of a web browser window showing the Jenkins login page. The browser's address bar displays the URL `103.250.10.251:9090/login?from=%2F` and indicates it is "Not secure". The page features the Jenkins logo on the left and a login form on the right. The form includes fields for "Username" (containing "devops") and "Password" (masked with dots), a "Keep me signed in" checkbox, and a blue "Sign in" button.

Sign in [Jenkins]

Not secure | 103.250.10.251:9090/login?from=%2F

Sign in to Jenkins

Username

devops

Password

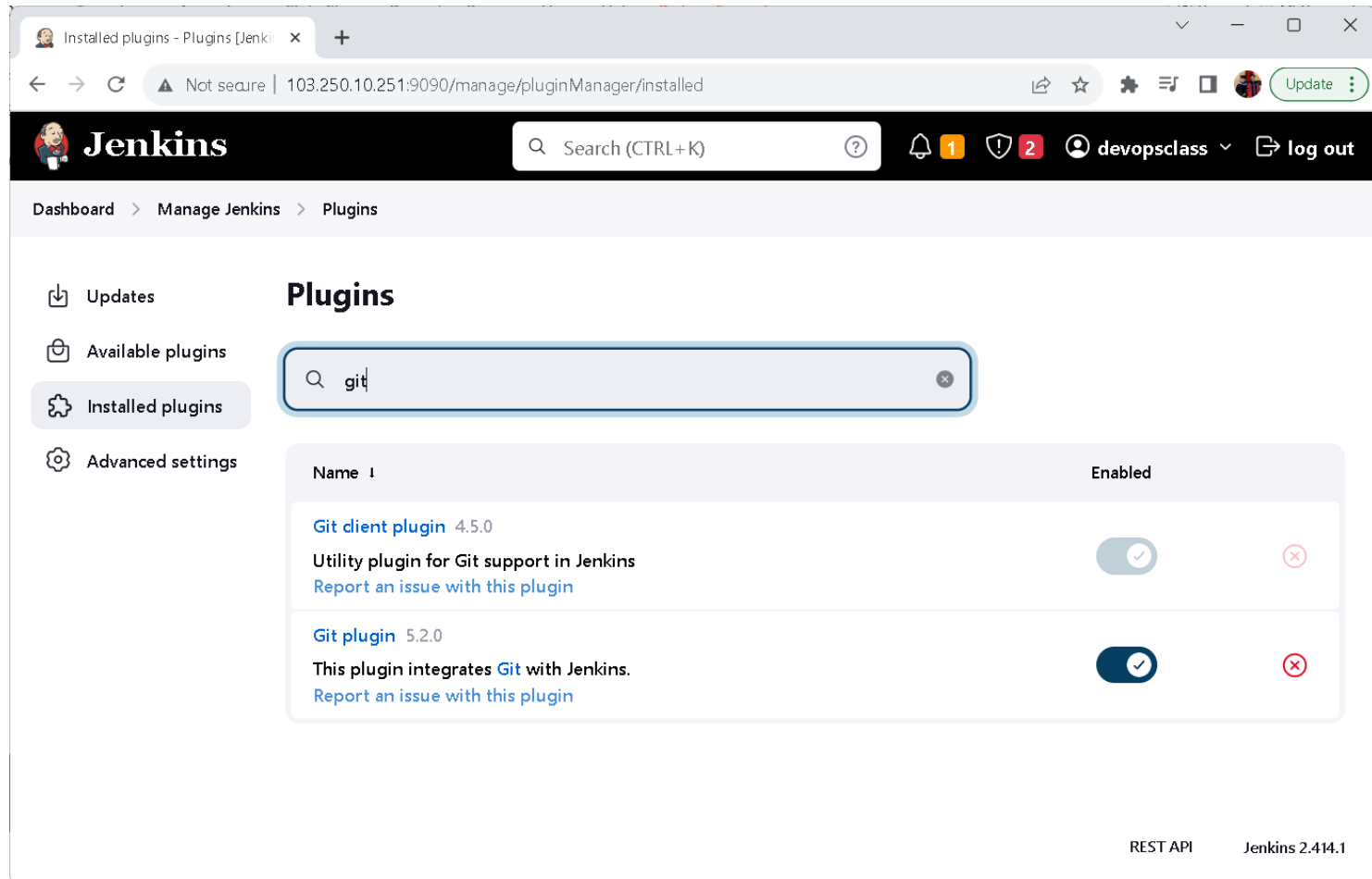
.....

☐ Keep me signed in

Sign in

Check Installed Plugins

Redirect mouse ke **Dashboard** > **"Manage Jenkins"** > **"Plugins"**.
Selanjutnya pilih tab **"Installed Plugins"**, cek tanda **ENABLED** seperti di bawah ini.



Installed plugins - Plugins [Jenki] x +

Not secure | 103.250.10.251:9090/manage/pluginManager/installed

Jenkins Search (CTRL+K) ? 1 2 devopsclass log out

Dashboard > Manage Jenkins > Plugins

Updates

Available plugins

Installed plugins

Advanced settings

Plugins

git

Name	Enabled
Git client plugin 4.5.0 Utility plugin for Git support in Jenkins Report an issue with this plugin	☒
Git plugin 5.2.0 This plugin integrates Git with Jenkins. Report an issue with this plugin	☒

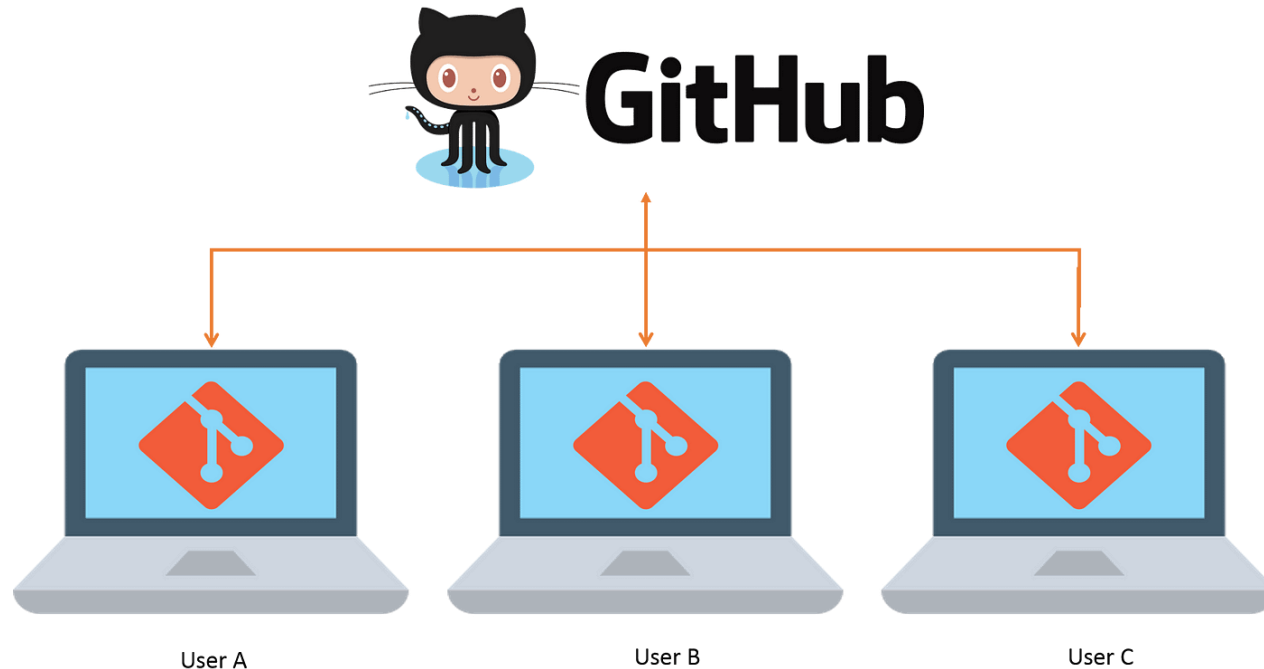
REST API Jenkins 2.414.1

[2]

Create SSH Public & Private Key Server



Jenkins



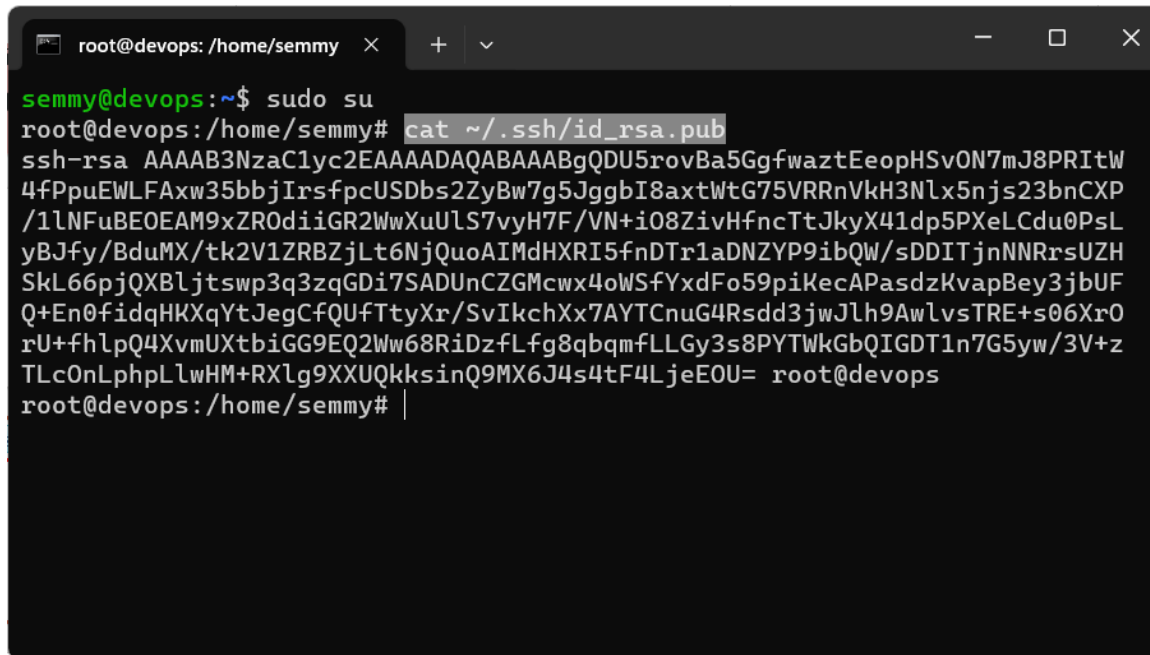
Mengapa harus menggunakan Git dan Github?

Akan digunakan sebagai Version Control, dimana akan hanya ada satu repo dalam project. Version control akan merekam semua perubahan yang terjadi, sehingga jika terjadi kesalahan, kita bisa dengan mudah kembali ke versi sebelumnya.

Check if SSH Public Key Exist?

Open Server di Putty atau CMD untuk login dan cek **PUBLIC KEY** SSH:

- **sudo su**
- **cat ~/.ssh/id_rsa.pub**

A terminal window with a dark background. The title bar shows 'root@devops: /home/semmy'. The prompt is 'semmy@devops:~\$'. The user enters 'sudo su', and the prompt changes to 'root@devops:/home/semmy#'. The user then enters 'cat ~/.ssh/id_rsa.pub'. The terminal displays a long string of text representing an SSH public key, starting with 'ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQgQDU5rovBa5GgfwaztEeopHSv0N7mJ8PRItW'. The output ends with 'root@devops: /home/semmy# |'.

Jika muncul **OUTPUT** seperti diatas, berarti public SSH key sudah ada. Kita akan hapus dulu private dan public SSH key ini dari CLOUD SERVER.

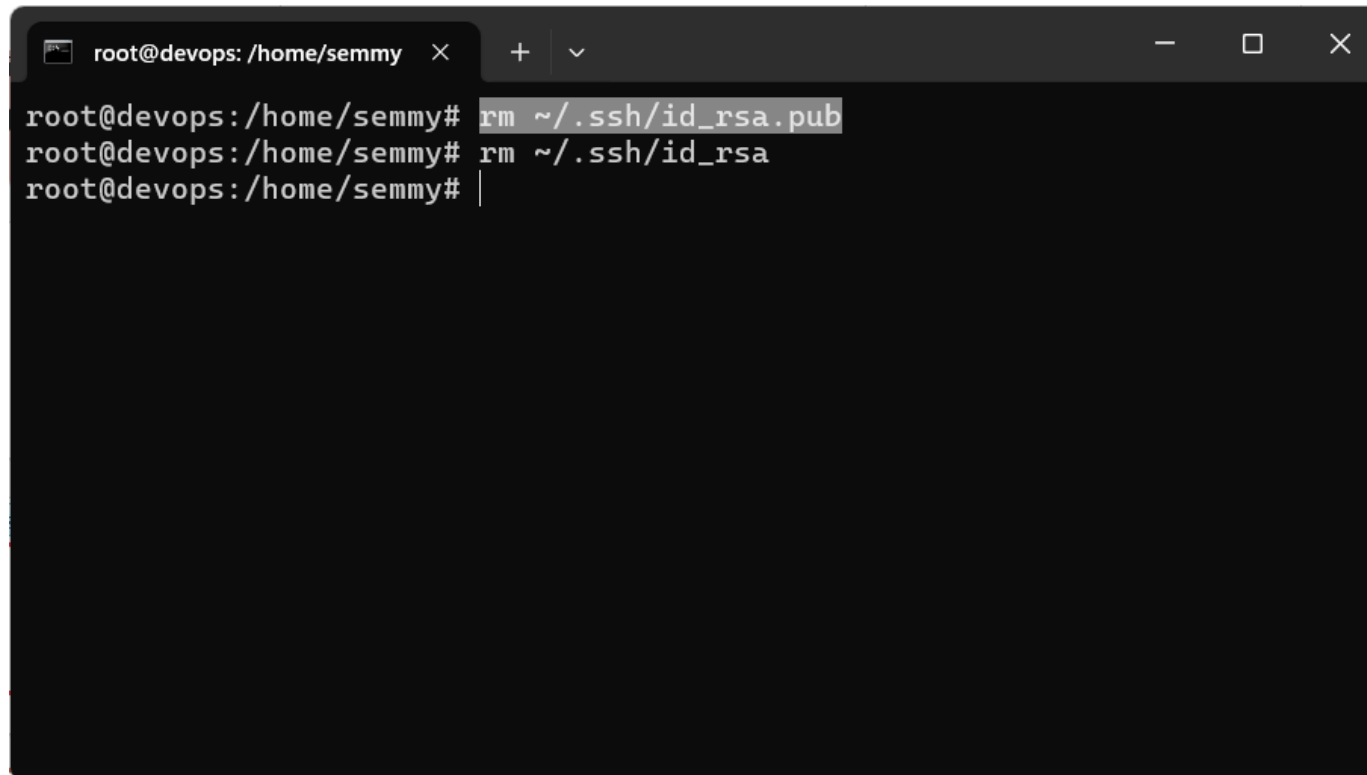
Delete Existing SSH Keys

Untuk menghapus SSH public key, kita dapat menjalankan perintah berikut:

➤ **rm ~/.ssh/id_rsa.pub**

Untuk menghapus SSH private key, kita dapat menjalankan perintah berikut:

➤ **rm ~/.ssh/id_rsa**

A terminal window with a dark background and light text. The window title bar shows 'root@devops: /home/semmy' and standard window controls. The terminal content shows three lines of commands and prompts: 'root@devops:/home/semmy# rm ~/.ssh/id_rsa.pub', 'root@devops:/home/semmy# rm ~/.ssh/id_rsa', and 'root@devops:/home/semmy# |'.

```
root@devops: /home/semmy × + ▾  
root@devops:/home/semmy# rm ~/.ssh/id_rsa.pub  
root@devops:/home/semmy# rm ~/.ssh/id_rsa  
root@devops:/home/semmy# |
```

Generating New SSH Key

Untuk membuat SSH key baru, kita dapat menggunakan perintah **ssh-keygen**. Ini akan menghasilkan sepasang kunci baru (*public dan private key*) dengan algoritma default, yaitu RSA.

[1] Jalankan perintah berikut dan tekan Enter:

➤ **ssh-keygen**

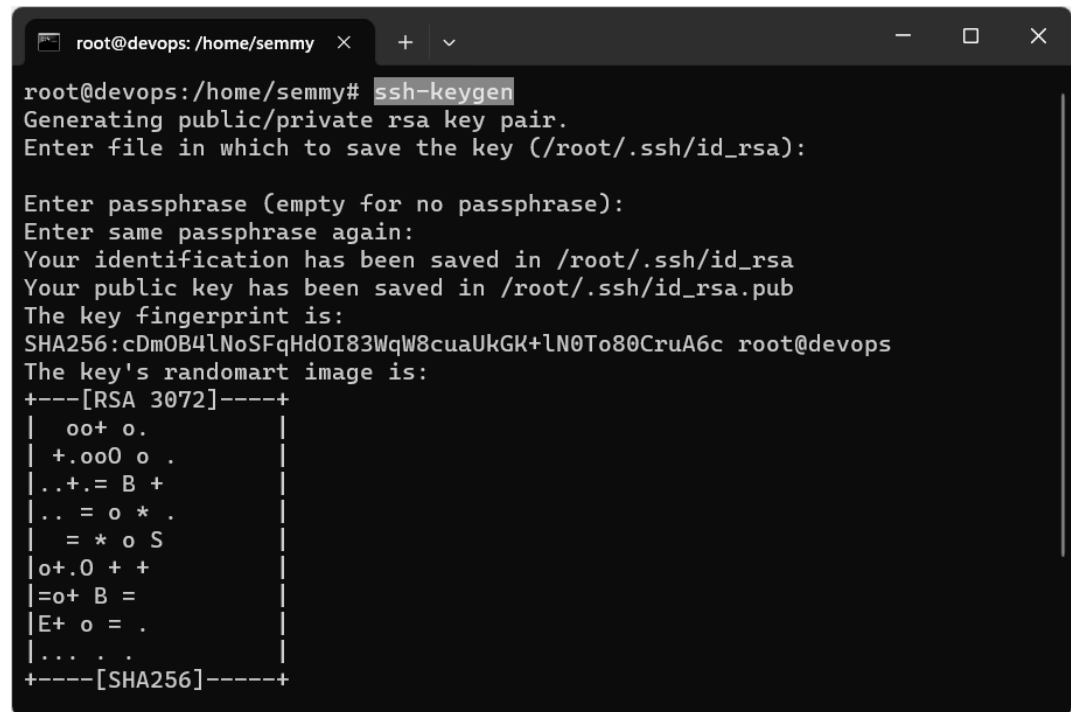
[2] Note, kita akan diminta untuk menentukan lokasi di mana untuk menyimpan SSH key yang baru. **Gunakan default lokasi** yang kunci akan disimpan di `~/.ssh/id_rsa` (*kunci privat*) dan `~/.ssh/id_rsa.pub` (*kunci publik*).

➤ Click ENTER

[3] **PASSPHRASE** dibiarkan kosong saja. **Pengalaman di dalam class, passphrase dapat menyebabkan loading forever**. Setelah selesai, kita akan memiliki kunci SSH yang baru.

Note:

Kita dapat memasukkan **PASSPHRASE** untuk melindungi private key kita.



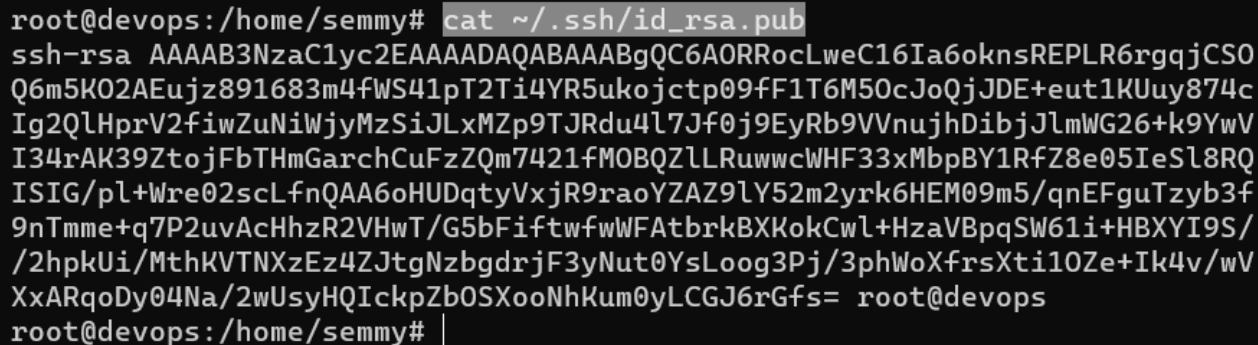
```
root@devops: /home/semmy  x  +  v
root@devops:/home/semmy# ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/root/.ssh/id_rsa):

Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /root/.ssh/id_rsa
Your public key has been saved in /root/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:cDmOB4lNoSFqHdOI83WqW8cuaUkGK+ln0To80CruA6c root@devops
The key's randomart image is:
+---[RSA 3072]-----+
|  oo+ o. |
| +.oo0 o . |
| ..+. = B + |
| .. = o * . |
|  = * o S |
| o+.0 + + |
| =o+ B = |
| E+ o = . |
| ... . . |
+-----[SHA256]-----+
```

Display New SSH Public Key

Kita dapat menampilkan kunci publik baru dengan perintah:

➤ **cat ~/.ssh/id_rsa.pub**

A terminal window with a dark background and light text. The title bar shows 'root@devops: /home/semmy' with window control buttons. The prompt is 'root@devops: /home/semmy#'. The command 'cat ~/.ssh/id_rsa.pub' is entered and executed. The output is a long string of characters representing the SSH public key, starting with 'ssh-rsa' and ending with 'root@devops'.

```
root@devops: /home/semmy# cat ~/.ssh/id_rsa.pub
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQGC6AORRocLWeC16Ia6oknsREPLR6rgqjCSO
Q6m5K02AEujz891683m4fWS41pT2Ti4YR5ukojctp09fF1T6M50cJoQjJDE+eut1KUuy874c
Ig2QlHprV2fiwZuNiWjyMzSiJLxMZp9TJRdu4l7Jf0j9EyRb9VVnujhDibjJlmWG26+k9YwV
I34rAK39ZtojFbTHmGarchCuFzZQm7421fMOBQZLLRuwwcWHF33xMbpBY1RfZ8e05IeSl8RQ
ISIG/pl+Wre02scLfnQAA6oHUDqtyVxjR9raoYZAZ9lY52m2yrk6HEM09m5/qnEFguTzyb3f
9nTmme+q7P2uvAcHhzR2VHwT/G5bFiftwfwWFAtbrkBXKokCwl+HzaVBpqSW61i+HBXYI9S/
/2hpkUi/MthKVTNXzEz4ZJtgNzbgrjF3yNut0YsLoog3Pj/3phWoXfrsXti10Ze+Ik4v/wV
XxARqoDy04Na/2wUsyHQIckpZb0SXooNhKum0yLCGJ6rGfs= root@devops
root@devops: /home/semmy# |
```

[3]

Setup SSH *in* Github



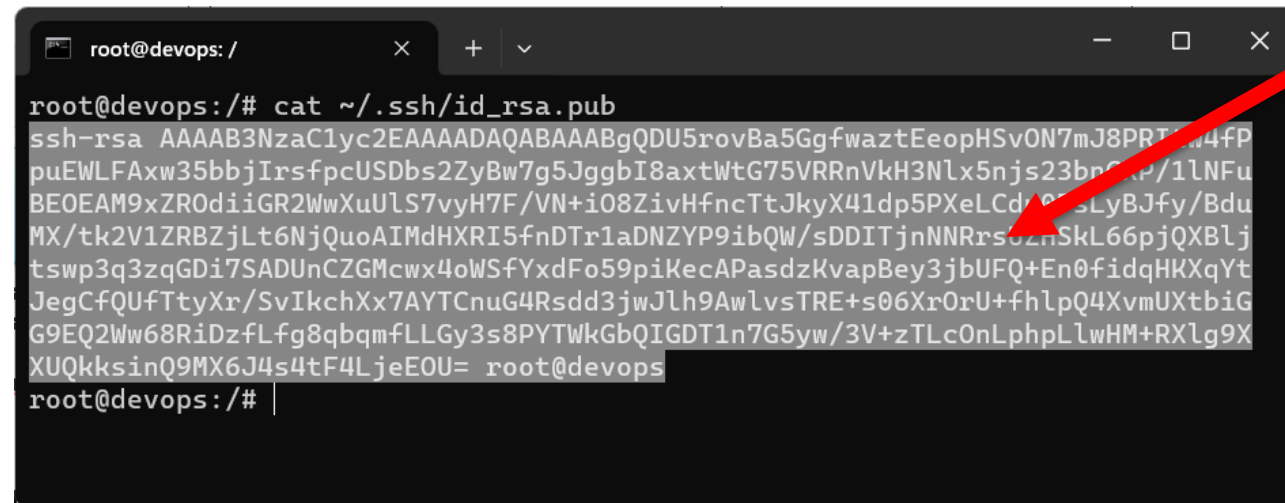
Jenkins

Setup SSH in Github

- ❖ Kita akan menggunakan Github sebagai Git Repository untuk menyimpan contoh project kita.
- ❖ Kita perlu melakukan setup Github terlebih dahulu, agar Jenkins bisa mengambil kode program di Github tanpa Login.
- ❖ Silahkan tambahkan SSH Public Key di VM Jenkins ke Github.

Open Server di Putty atau CMD untuk login dan dapat **PUBLIC KEY SSH:**

- sudo su
- cat ~/.ssh/id_rsa.pub

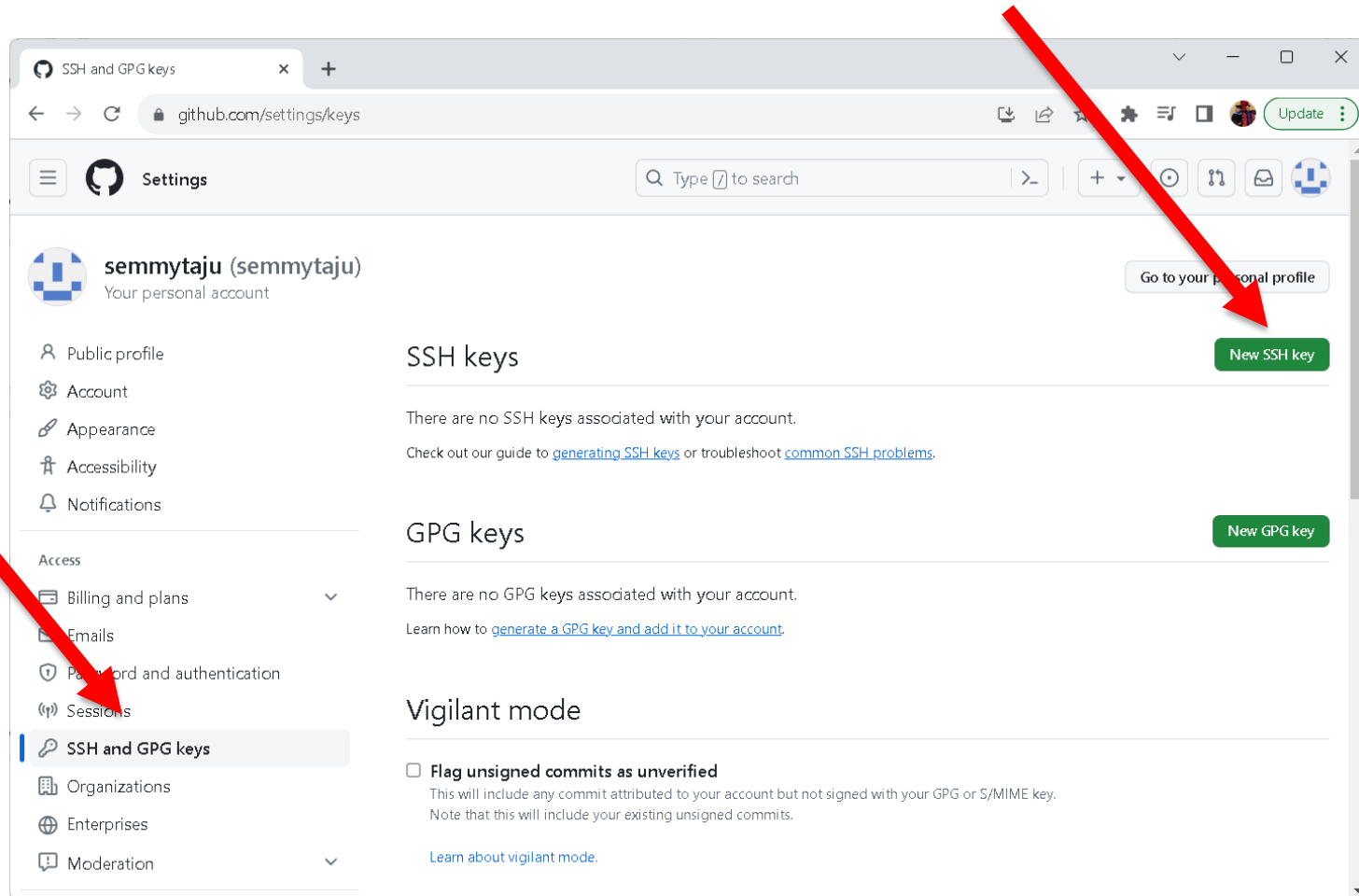
A terminal window titled 'root@devops: /' with standard window controls. The command 'cat ~/.ssh/id_rsa.pub' has been executed, and the output is a long string of alphanumeric characters representing the SSH public key. A red arrow points from the text 'Select dan copy public key.' to the output text.

```
root@devops:/# cat ~/.ssh/id_rsa.pub
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQGDQU5rovBa5GgfWaztEeopHSvON7mJ8PRTm4fP
puEWLFAxw35bbjIrsfpcUSDbs2ZyBw7g5JggBI8axtWtG75VRRnVkh3Nlx5njs23bnGAP/1lNFu
BEOEAM9xZR0diiGR2WwXuUls7vyH7F/VN+i08ZivHfncTtJkyX41dp5PXeLCd093LyBJfy/Bdu
MX/tk2V1ZRBZjLt6NjQuoAIMdHXRI5fnDTr1aDNZYP9ibQW/sDDITjnNNRrs0znSkL66pjQXBlj
tswp3q3zqGDi7SADUnCZGMcwx4oWSfYxdFo59piKecAPasdzKvapBey3jbUFQ+En0fidqHKXqYt
JegCfQUfTtyXr/SvIkchXx7AYTCnuG4Rsdd3jwJlh9AwlvTRE+s06XrOrU+fhlpQ4XvmUXtbiG
G9EQ2Ww68RiDzfLfg8qbqmFLLGy3s8PYTWkGbQIGDT1n7G5yw/3V+zTLC0nLphpLlwHM+RXlg9X
XUQkksinQ9MX6J4s4tF4LjeE0U= root@devops
root@devops:/#
```

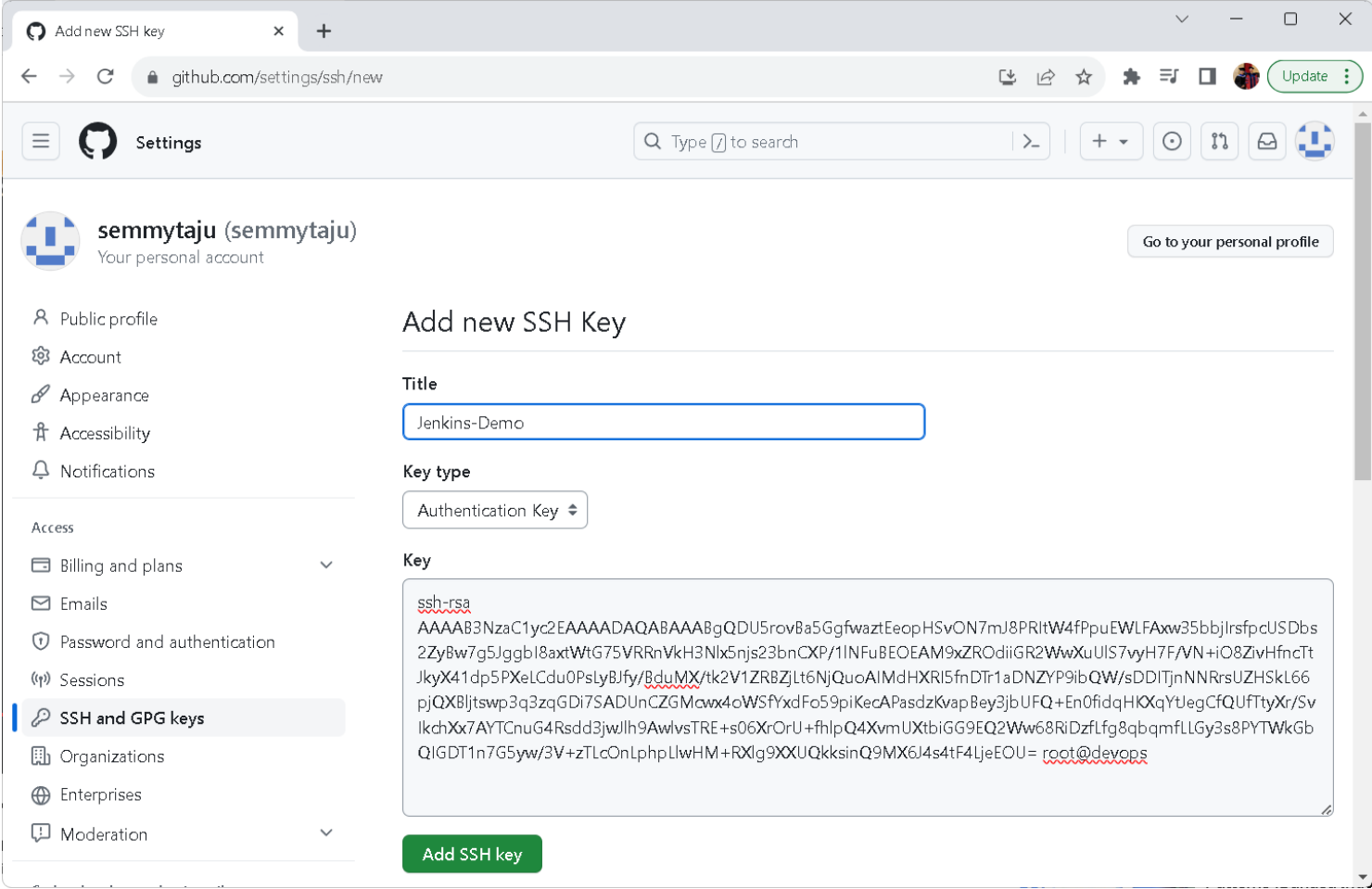
**Select dan
copy public
key.**

Add SSH keys *in* Github

Masuk ke Github, arahkan ke **Profile** dan pilih **Settings**. Disamping kiri pilih **SSH and GPG keys**. Click button “New SSH key” untuk menambahkan SSH key yang baru.



Add New SSH Public Key



Add new SSH key

github.com/settings/ssh/new

Settings

semmytaju (semmytaju)
Your personal account

Go to your personal profile

Public profile
Account
Appearance
Accessibility
Notifications

Access

Billing and plans
Emails
Password and authentication
Sessions
SSH and GPG keys
Organizations
Enterprises
Moderation

Add new SSH Key

Title

Jenkins-Demo

Key type

Authentication Key

Key

```
ssh-rsa
AAAAB3NzaC1yc2EAAAADAQABAAQBAgQDU5rovBa5GgfwaztEeopHSvON7mJ8PRitW4fPpuEwLFAxw35bbjIrsfpcUSDbs
2Zy8w7g5JggbI8axtWtG75VRRnVkh3Nlx5njs23bnCXP/1INFuBEOEAM9xZROdiiGR2WwXuUIS7vyH7F/VN+iO8ZivHfncTt
JkyX41dp5PXeLCdu0PsLyBJfy/BduMX/tk2V1ZR8ZjLt6NjQuoAIMdHXRi5fnDTr1aDNZYP9ibQW/sDDITjnNNRsUZHskL66
pjQX8Bljtswp3q3zqGDt7SADUnCZGMowx4oWSFYxdFo59piKecAPasdzKvapBey3jbUFQ+En0fidqHXKqYUegCfQUftTyXr/Sv
lkchXx7AYTCnuG4Rsdd3jwlh9AwlvstRE+s06XrOrU+fhlpQ4XvmUXtbiGG9EQ2Ww68RiDzflfg8qbqmfLLGy3s8PYTWkGb
QIGDT1n7G5yww/3V+zTlCOnLphlUwHM+RXlg9XXUQkksinQ9MX6J4s4tF4ljeEOU= root@devops
```

Add SSH key

New Authentication Keys Generated

The screenshot shows the GitHub 'SSH and GPG keys' settings page for the user 'semmytaju'. A blue banner at the top of the page content area states: 'You have successfully added the key 'Jenkins-Demo''. Below this, the user's profile is shown with the name 'semmytaju (semmytaju)' and the text 'Your personal account'. To the right of the profile is a button that says 'Go to your personal profile'. On the left side, there is a sidebar menu with options: 'Public profile', 'Account', 'Appearance', 'Accessibility', 'Notifications', 'Access', 'Billing and plans', 'Emails', and 'Password and authentication'. The main content area is titled 'SSH keys' and has a green button labeled 'New SSH key'. Below the title, it says: 'This is a list of SSH keys associated with your account. Remove any keys that you do not recognize.' Under the heading 'Authentication Keys', there is a single key entry for 'Jenkins-Demo'. This entry includes a key icon, the key name, the SHA256 fingerprint 'zIVB8vE7d20PxLWktEY/zgScZXYJSUFR8bSxFydSsCQ', the text 'Added on Oct 23, 2023', and the permissions 'Never used — Read/write'. A 'Delete' button is located to the right of the key entry. At the bottom of the main content area, there is a link to a guide: 'Check out our guide to [generating SSH keys](#) or troubleshoot [common SSH problems](#).'

SSH and GPG keys

github.com/settings/keys

Settings

You have successfully added the key 'Jenkins-Demo'.

semmytaju (semmytaju)
Your personal account

Go to your personal profile

Public profile
Account
Appearance
Accessibility
Notifications

SSH keys

New SSH key

This is a list of SSH keys associated with your account. Remove any keys that you do not recognize.

Authentication Keys

Jenkins-Demo
SHA256: zIVB8vE7d20PxLWktEY/zgScZXYJSUFR8bSxFydSsCQ
Added on Oct 23, 2023
Never used — Read/write

Delete

Check out our guide to [generating SSH keys](#) or troubleshoot [common SSH problems](#).

[4]

Configure Credentials Jenkins



Jenkins

Credentials Jenkins

Konfigurasi kredensial (*credentials*) di Jenkins digunakan untuk menyimpan dan mengelola informasi autentikasi yang diperlukan oleh Jenkins dalam berbagai skenario. Informasi autentikasi ini dapat mencakup *username* dan *password*, token, atau kunci API yang diperlukan oleh Jenkins untuk mengakses berbagai sumber daya, seperti repositori kode sumber, server database, layanan cloud, dan lain sebagainya.

Open Server di Putty atau CMD untuk login dan dapat SSH **PRIVATE KEY** :

- **sudo su**
- **cat ~/.ssh/id_rsa**

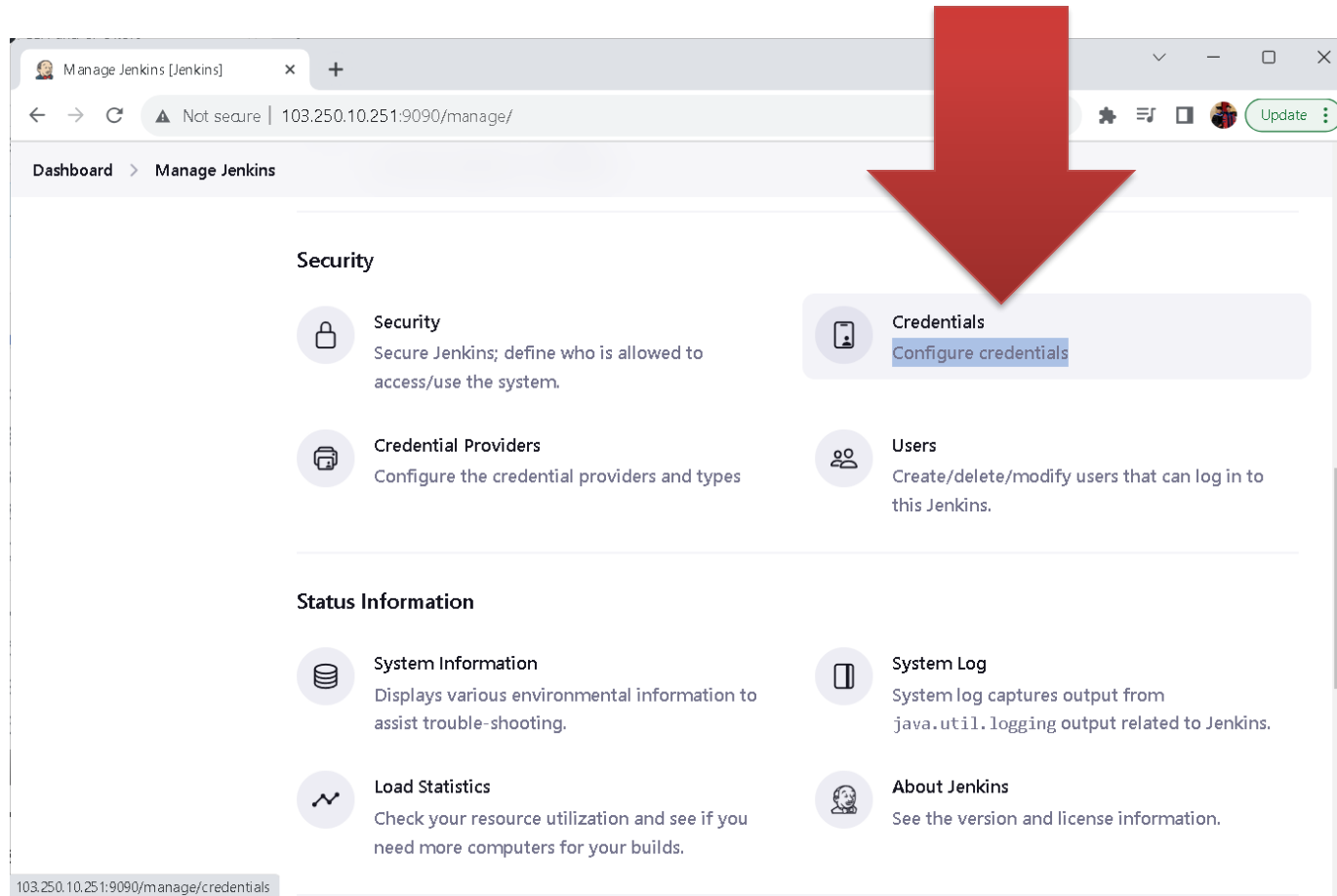


```
root@devops: /
root@devops: /# cat ~/.ssh/id_rsa
-----BEGIN OPENSSH PRIVATE KEY-----
b3BlbnNzaC1rZXktdjEAAAACmFlczI1Ni1jdHIAAAAGYmNyeXB0AAAAGAAAABADtiezam
z/Z6R3lQfLefPAAAAEAAAAEAAAGXAAAAB3NzaC1yc2EAAAADAQABAAQgQDU5rovBa5G
gfwaztEeopHSvON7mJ8PRIW4fPpuEwLFAxw35bbjIrsfpcUSDBs2ZyBw7g5JggBI8axtW
tG75VRRnVkh3Nlx5njs23bnCXP/1lNFuBE0EAM9xZR0diGR2WwXuUls7vyH7F/VN+i08Z
ivHfncTtJkyX41dp5PXeLCdu0PsLyBjfy/BduMX/tk2V1ZRBZjLt6NjQuoAIMdHXRI5fnD
Tr1aDNZYP9ibQW/sDDITjnNNRrsUZHSkL66pjQXBljtswp3q3zqGDi7SADUnCZGMcw4oW
SfYxdFo59piKecAPasdzKvapBey3jBUfQ+En0fidqHKXqYtJegCfQUfTtyXr/SvIkchXx7
AYTCnuG4Rsdd3jwJlh9AwlvsTRE+s06XrOrU+fhlpQ4XvmUXtbiGG9EQ2Ww68RiDzfLfg8
qbqmfLLGy3s8PYTWkGbQIGDT1n7G5yw/3V+zTLcOnLphpLlwHM+RXlg9XXUQkksinQ9MX6
J4s4tF4LjeE0UAAAWAhV2baRvaerXuHxK6Vr/w+wWaS6BRkLWxvmszLXOC6rJELNkdEcQR
+qpEQBW0qJiwPxy6Z6xtrU0wn7UbbJpBoHES/JITBlCPK7tHBMK+4K60BdserjcpCh+CD
R5fL5H5A2YXIEj2XtbUE+7oWAwinaRlFNGezdvErHRuhm1XTHXQ+HKGuHrm1lvStVTLrBQ
YZUC2UsAVBW7f2BRxBy/ahExEdmQA0EGL1KVgrfzBZCKnSk0g90Hd3hj1ntvttkkZYfoZg
CDnhh5r2Rt3561SU2Tsk15xQZ1IjZGYcobPUZZ/Si2Z4+Kv0k6tz7KPMwVIAicAiSkLI
tnaZKZ6aPHnlvdb8EUIutwBxo0BA0Ktps6iDW53W7QUdk4oDYWq28Ly2Zw7b4ziQrMwbXR
MaE+I1DlQBGdyMqIjabQRNByZJq0Z58MjWhKtb6Y15zbPDKK/OKJo4WFonU2zy2DPZABGI
tfJ+YbZQut6UPS932LOpoP7qIA2DmX9XkTD3/AAUZIjaLb5TEpYucTry6FDD399WGucA3L
en7tMO/oqam1iivtdKyTymJXh5+nIOlqltwkmYm3vVs89HqBEnIwqiXHxkHbLhxKgPoPfOW
BDVATBIFW1FnG36iD1vDmMep0RP+hgLJZENsVXGJxNBq6uThD9VnF1GrfCf2z1pLV3pxx
nqwyqlCV5Z+HXLpVowF2BKcTsaUmbxLgNe4n2TX8Rye0jacpIa/n1yU9+j0+mHgQLaGWY/
```

Select dan
copy private
ssh key.

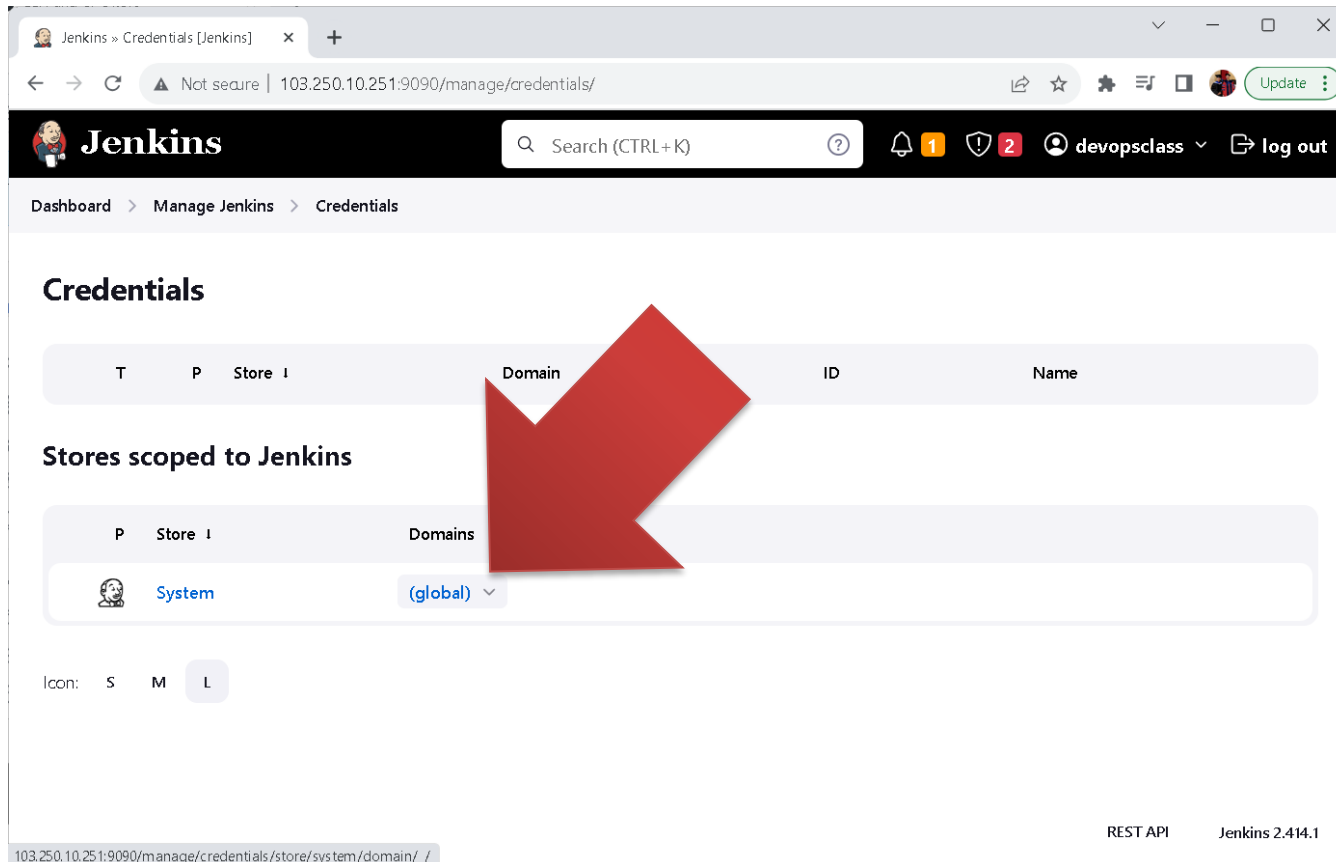
Global Credentials Jenkins (Unrestricted)

Redirect mouse ke **Dashboard** > "**Manage Jenkins**" > "**Credentials**".



Global Credentials Jenkins

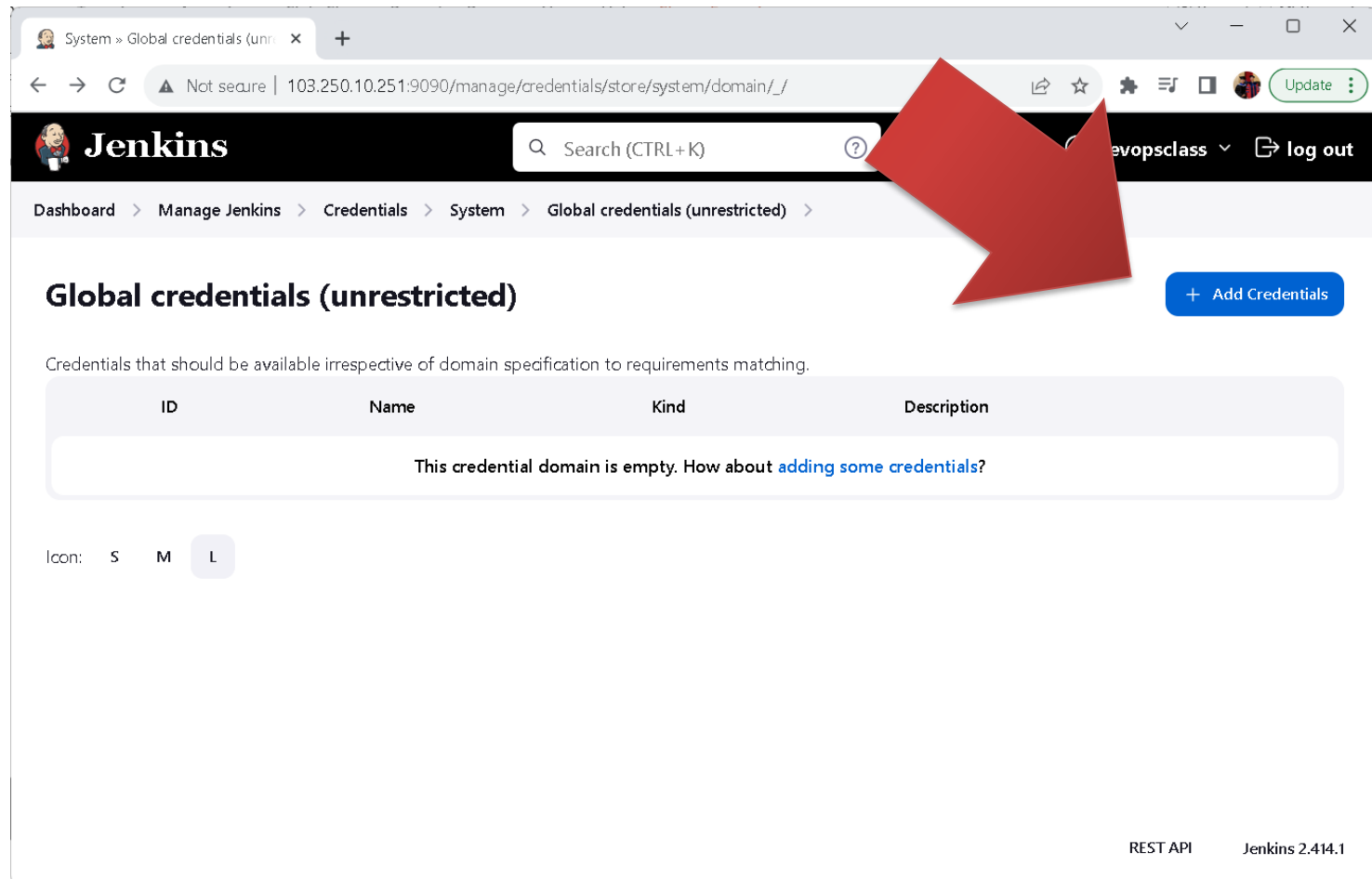
Pilih “**global**” untuk membuat global credentials Jenkins (unrestricted).



The screenshot shows the Jenkins web interface at the URL `103.250.10.251:9090/manage/credentials/`. The page title is "Credentials". Below the title, there is a table with columns: T, P, Store, ID, Domain, and Name. The table is currently empty. Below the table, there is a section titled "Stores scoped to Jenkins". This section contains a table with columns: P, Store, and Domains. The first row in this table shows a store named "System" with a dropdown menu for "Domains" set to "(global)". A large red arrow points to the "(global)" dropdown menu. At the bottom of the page, there is a footer with the text "REST API" and "Jenkins 2.414.1".

Add New Credentials

Click button “**Add Credentials**” untuk membuat global credentials Jenkins.



The screenshot shows the Jenkins web interface. The browser address bar indicates the URL is `103.250.10.251:9090/manage/credentials/store/system/domain/_/`. The Jenkins logo and a search bar are at the top. The breadcrumb trail is: `Dashboard > Manage Jenkins > Credentials > System > Global credentials (unrestricted)`. The main heading is **Global credentials (unrestricted)**. Below it, a text line says: "Credentials that should be available irrespective of domain specification to requirements matching." A table with columns **ID**, **Name**, **Kind**, and **Description** is shown, but it is empty. A message inside the table area says: "This credential domain is empty. How about [adding some credentials?](#)". A blue button labeled **+ Add Credentials** is in the top right corner of the main content area. A large red arrow points from the right side of the page towards this button. At the bottom right, the text **REST API** and **Jenkins 2.414.1** is visible.

New Credentials

New credentials [Jenkins] x +

Not secure | 103.250.10.251:9090/manage/credentials/store/system/domain/_/newCredentials

Jenkins Search (CTRL+K) ? 1 2 devopsclass log out

Dashboard > Manage Jenkins > Credentials > System > Global credentials (unrestricted) >

New credentials

Kind

SSH Username with private key

Scope ?

Global (Jenkins, nodes, items, all child items, etc)

ID ?

github-semmytaju

Description ?

Github Semmy

Username

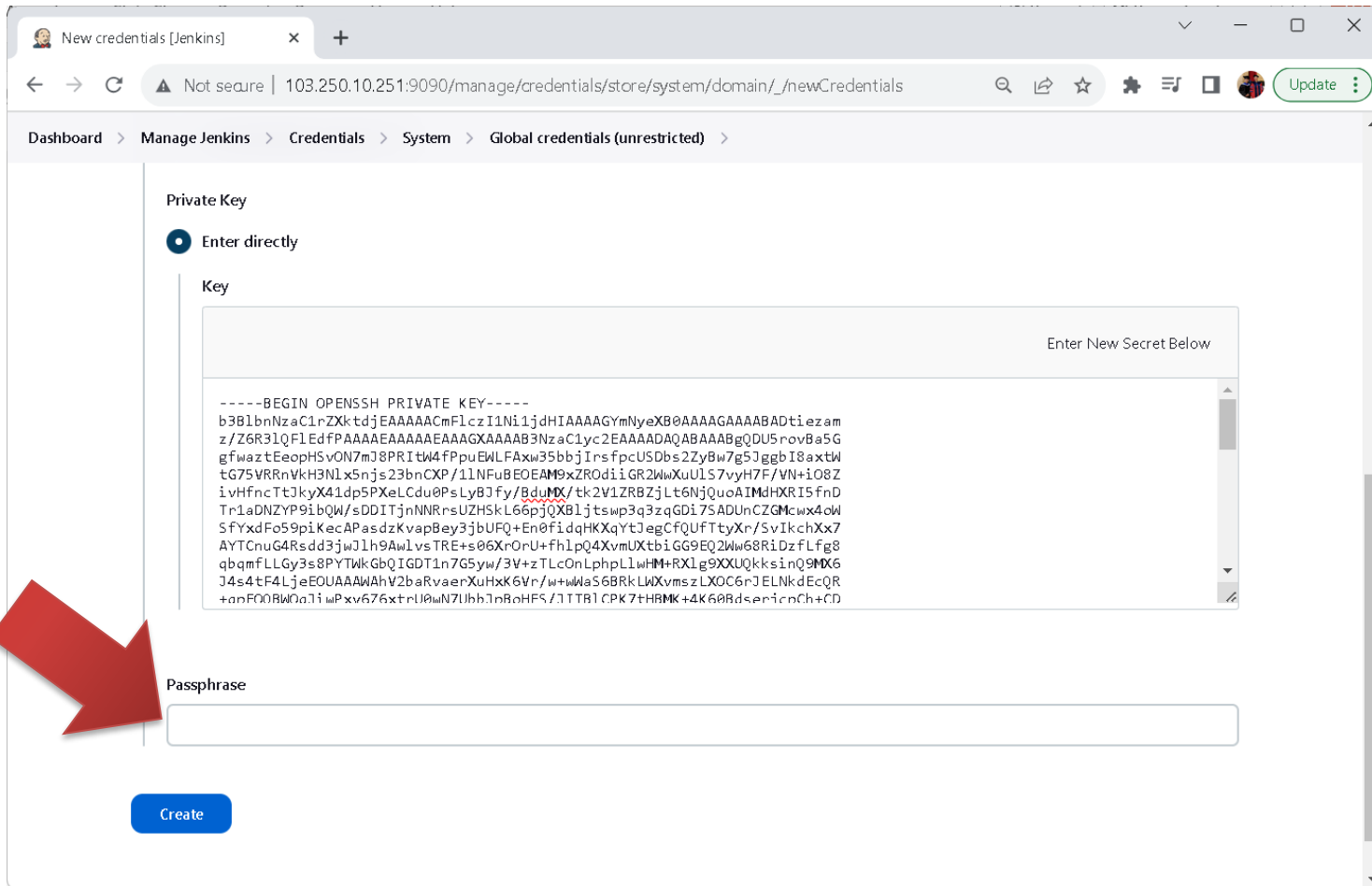
semmytaju

Create

Optional

Optional

Add Private Key to Credentials



The screenshot shows the Jenkins 'New credentials' page. The breadcrumb trail is: Dashboard > Manage Jenkins > Credentials > System > Global credentials (unrestricted). The 'Private Key' section is active, with the 'Enter directly' radio button selected. The 'Key' field contains a long, multi-line SSH private key. The 'Passphrase' field is empty. A red arrow points to the 'Passphrase' field. The 'Create' button is at the bottom left.

Private Key

☒ Enter directly

Key

Enter New Secret Below

```
-----BEGIN OPENSSH PRIVATE KEY-----
b3B1bnNzaC1rZXktdjEAAAACmFlczI1Ni1jdHIAAAAGYmNyeXB0AAAAAGAAABADtiezam
z/Z6R31QF1EdfPAAAAEAAAAEAAAGXAAAAB3NzaC1yc2EAAAADAQABAAQgQDU5rovBa5G
gfwaztEeopHSvON7mJ8PRItW4fPpuEWLFAxw35bbjIrsfpcUSDbs2ZyBw7g5Jggbi8axtW
tG75VRRnVkh3N1x5njs23bnCXP/11NFuBEOEAM9xZROdiGR2WwXuU1S7vyH7F/VN+i08Z
ivHfncTtJkyX41dp5PXeLCdu0PsLyBjfy/BduMX/tk2V1ZRBZjLt6NjQuoAIMdHXRI5fnD
Tr1aDNZYP9ibQW/sDDITjnnNRrsUZHskL66pjQXB1jtswp3q3zqGDI7SADUnCZGMcwx4oW
SFYxdFo59piKecAPasdzKvapBey3jbUFQ+En0fidqHKXqYtJegCFQUFTtyXr/SvIkchXx7
AYTCnuG4Rsdd3jwJ1h9Aw1vsTRE+s06XrOrU+fhlpQ4XvmUXtbiGG9EQ2Ww68RiDzFLfg8
qbqmFLGy3s8PYTWkGbQIGDT1n7G5yw/3V+zTLcOnLphpLLwHM+RX1g9XXUQkksinQ9MX6
J4s4tF4LjeEOUAAAHV2baRvaerXuHxK6Vr/w+uW56BRkLW4XvmszLXOC6nJELNkdEcQR
+onFO0BWDnJiwpXv676xtU0wN7UbbJnBoHFS/1TTB1CPK7tHBMK+4K60BdsericpCh+CD
```

Passphrase

Create

Perhatian:

PASSPHRASE dibiarkan kosong saja, sesuaikan dengan step awal pada saat SSH dibuat. Pengalaman di dalam class, passphrase dapat menyebabkan loading forever.

Global Credentials Created

The screenshot shows the Jenkins web interface. The browser address bar indicates the URL `103.250.10.251:9090/manage/credentials/store/system/domain/_/`. The Jenkins header includes the logo, a search bar, and user information for 'devopsclass'. The breadcrumb trail is: Dashboard > Manage Jenkins > Credentials > System > Global credentials (unrestricted). The main heading is 'Global credentials (unrestricted)' with an '+ Add Credentials' button. Below the heading is a note: 'Credentials that should be available irrespective of domain specification to requirements matching.' A table lists the credentials:

ID	Name	Kind	Description
 github-semmytaju	semmytaju (Github Semmy)	SSH Username with private key	Github Semmy 

Below the table, there are icons for 'Icon: S M L'. At the bottom right, it says 'REST API' and 'Jenkins 2.414.1'.

[5]

Integrasi Git dan Jenkins

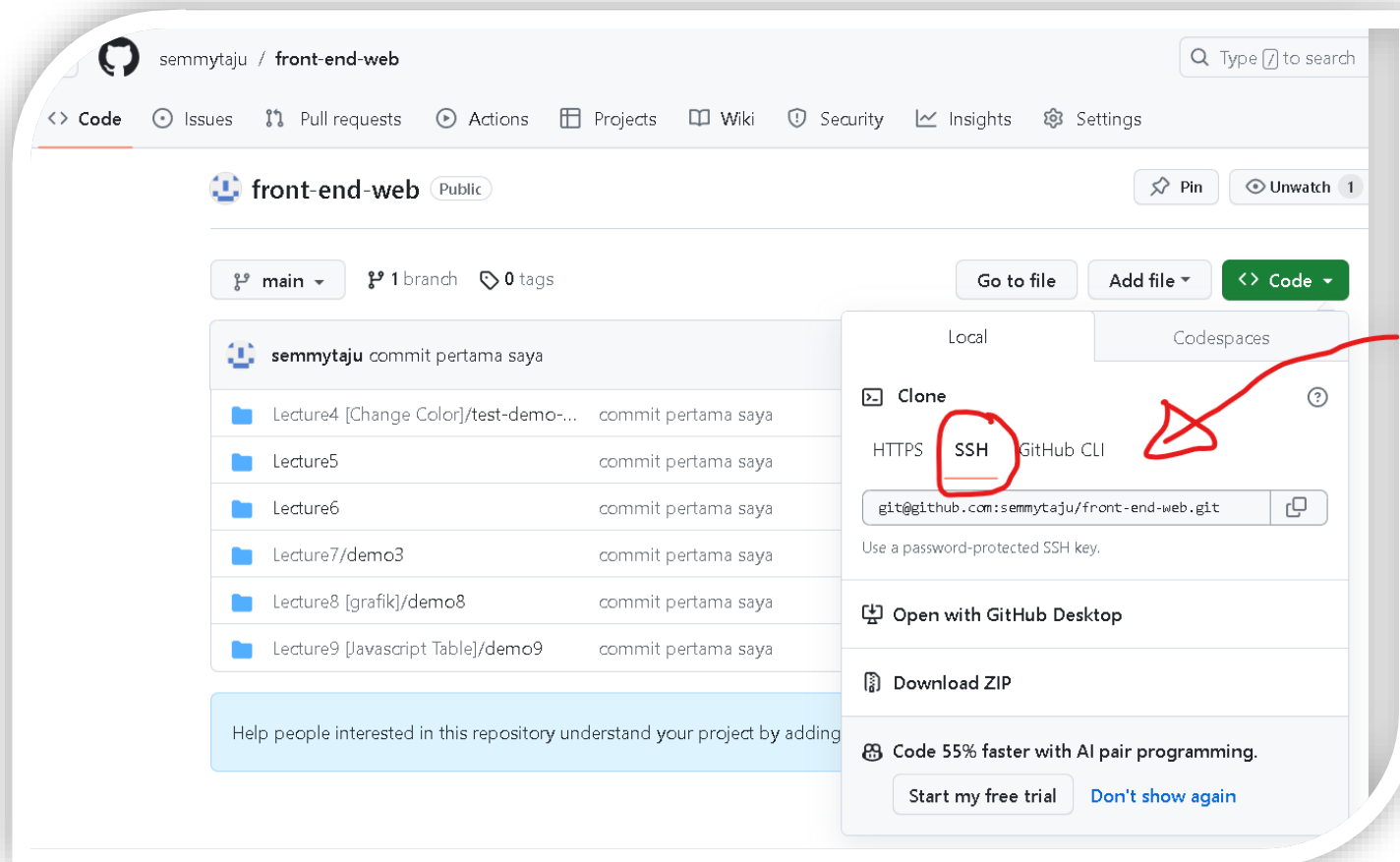


Jenkins

Contoh Project di Github

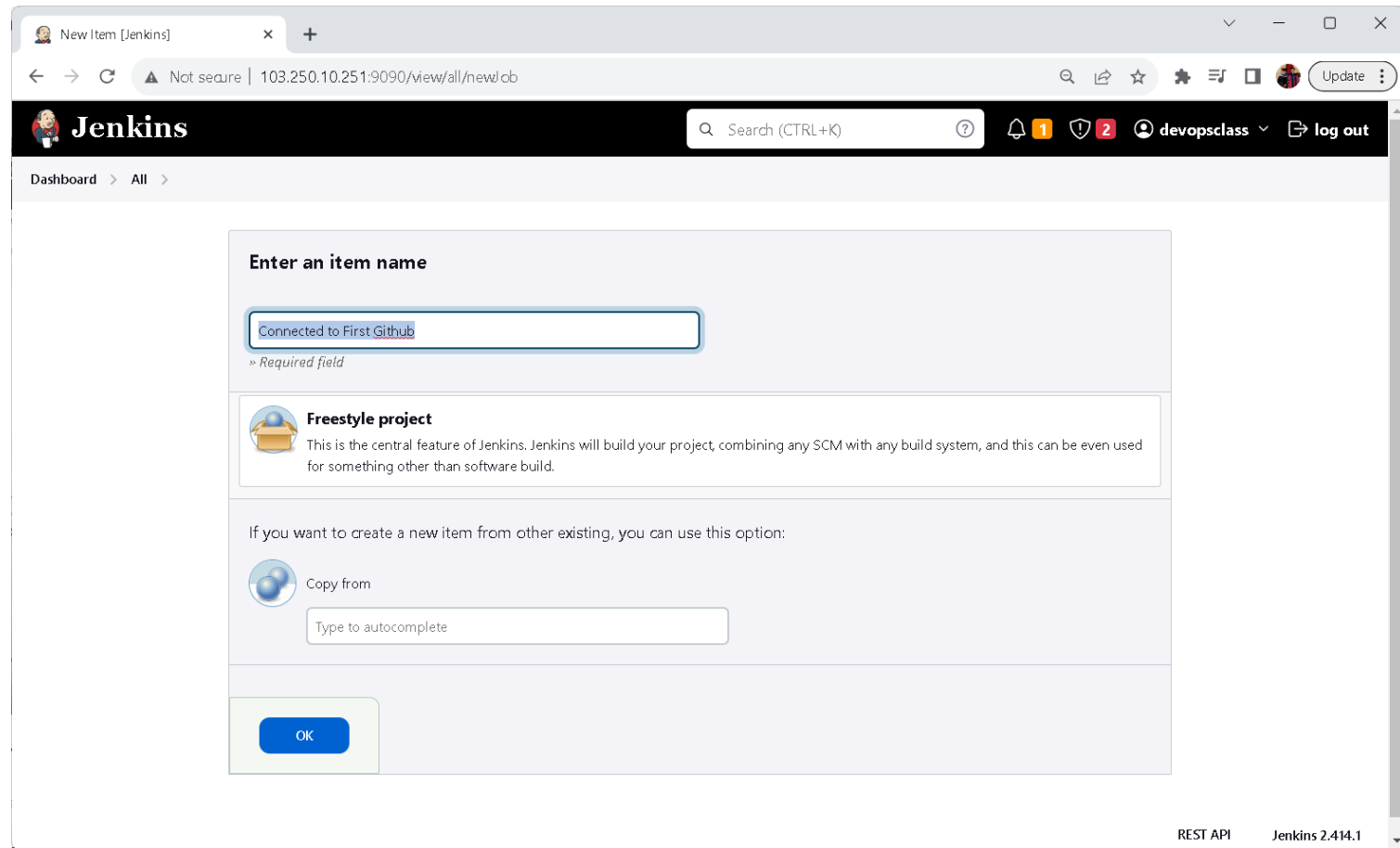
SSH Link Github:

<git@github.com:semmytaju/front-end-web.git>



Create New Jenkins Job

Buat sebuah Job baru dengan nama **“Connected to First Github”**.



The screenshot shows the Jenkins web interface in a browser window. The title bar indicates 'New Item [Jenkins]'. The address bar shows '103.250.10.251:9090/view/all/newJob'. The Jenkins logo and 'Jenkins' text are in the top left. A search bar with 'Search (CTRL+K)' is in the top right. Below the header, there's a breadcrumb 'Dashboard > All >'. The main content area has a section titled 'Enter an item name' with a text input field containing 'Connected to First Github'. Below the input field is a red asterisk and the text '» Required field'. Underneath is a 'Freestyle project' option with a description: 'This is the central feature of Jenkins. Jenkins will build your project, combining any SCM with any build system, and this can be even used for something other than software build.' Below that, there's a section 'If you want to create a new item from other existing, you can use this option:' with a 'Copy from' option and a text input field with the placeholder 'Type to autocomplete'. At the bottom left of the form is a blue 'OK' button. The bottom right of the page shows 'REST API' and 'Jenkins 2.414.1'.

New Item [Jenkins]

Not secure | 103.250.10.251:9090/view/all/newJob

Jenkins

Search (CTRL+K)

devopsclass log out

Dashboard > All >

Enter an item name

Connected to First Github

» Required field

Freestyle project
This is the central feature of Jenkins. Jenkins will build your project, combining any SCM with any build system, and this can be even used for something other than software build.

If you want to create a new item from other existing, you can use this option:

Copy from

Type to autocomplete

OK

REST API Jenkins 2.414.1

Konfigurasi Git di Jenkins Job

The screenshot shows the Jenkins web interface in a browser. The address bar indicates the URL is `103.250.10.251:9090/job/Connected%20to%20First%20Github/configure`. The Jenkins logo and a search bar are at the top. The breadcrumb trail is `Dashboard > Connected to First Github > Configuration`. On the left, the 'Configure' menu is open, showing options: General (selected), Source Code Management, Build Triggers, Build Environment, Build Steps, and Post-build Actions. The 'General' tab is active, showing a toggle for 'Enabled' which is turned on. The 'Description' field contains the text 'Connect ke github repository.' Below this, there are three unchecked checkboxes: 'Discard old builds', 'This project is parameterized', and 'Execute concurrent builds if necessary'. An 'Advanced' dropdown menu is visible. At the bottom, the 'Source Code Management' section is partially visible, and there are 'Save' and 'Apply' buttons.

Connected to First Github Config

Not secure | 103.250.10.251:9090/job/Connected%20to%20First%20Github/configure

Jenkins

Search (CTRL+K)

devopsclass log out

Dashboard > Connected to First Github > Configuration

Configure

- General
- Source Code Management
- Build Triggers
- Build Environment
- Build Steps
- Post-build Actions

General Enabled

Description

Connect ke github repository.

Plain text [Preview](#)

☐ Discard old builds ?

☐ This project is parameterized ?

☐ Execute concurrent builds if necessary ?

Advanced

Source Code Management

Save Apply

Konfigurasi Git repository di Jenkins

Pada bagian **Source Code Management** di menu “**configure**” Job Jenkins. Sudah terdapat opsi untuk git.

Student Note:

Pilih **Credentials** nya dulu, selanjutnya mengisi field pada “**Repository URL**”.

The screenshot shows the Jenkins 'Configure' page for a job named 'Connected to First Github'. The 'Source Code Management' section is active, showing the 'Git' option selected. A red box highlights the 'Repository URL' field, which contains 'git@github.com:senmytaju/front-end-web.git'. Another red box highlights the 'Credentials' dropdown menu, which shows 'senmytaju (Github Senmy)' selected. A red arrow points from the 'Credentials' dropdown to a text box on the right. A large red arrow points from the bottom left towards the 'Source Code Management' section, with the text 'Start dari bawah ke atas.' below it. The number '2' is next to the 'Repository URL' field, and the number '1' is next to the 'Credentials' dropdown.

Start dari bawah ke atas.

2

1

Akan mengalami loading forever pada credential saat configures Job di Jenkins

Error: Failed to connect to repository

Jika kalian menghadapi tipe error yang sama pada point ke-2 di slide sebelumnya, errornya seperti di bawah ini:

Failed to connect to repository : Command "git ls-remote -h -- git@github.com:semmyTaju/excercise3.git HEAD" returned status code 128:stdout:stderr: No ED25519 host key is known for github.com and you have requested strict checking.**Host key verification failed.fatal: Could not read from remote repository.**Please make sure you have the correct access rightsand the repository exists.

Buka server menggunakan Putty atau CMD:

- sudo su
- ssh-keyscan github.com >> ~/.ssh/known_hosts

Menambahkan Build Step

Kita bisa melakukan **konfigurasi Job** untuk menambahkan **Build Step** yang kita mau di Jenkins.

The screenshot shows the Jenkins web interface. The browser address bar indicates the URL is `103.250.10.251:9090/job/Connected%20to%20First%20Github/configure`. The page title is "Configure" and the breadcrumb trail is "Dashboard > Connected to First Github > Configuration".

On the left sidebar, the "Build Steps" option is selected. The main content area shows the "Build Steps" configuration. A red rectangle highlights the "Add build step" button and its dropdown menu. The dropdown menu is open, showing three options: "Execute Windows batch command", "Execute shell" (which is highlighted in blue), and "Invoke top-level Maven targets".

Below the "Build Steps" section, there is an "Add post-build action" button and "Save" and "Apply" buttons at the bottom.

At the bottom right of the page, it says "REST API" and "Jenkins 2.414.1".

Otomasi Git Clone dari Github

Jadi yang akan dibuat adalah setiap 2 menit jenkins akan melakukan otomasi git clone dari repo github. Ikuti Langkah-langkah dibawah ini. Bisa juga mengubah semauanya untuk otomasi dilakukan setiap berapa waktu bisa menit jam hari dan sebagainya. Kemudian di save.



The screenshot shows the Jenkins Configuration page for a job named "Connection Github kedua". The breadcrumb trail at the top is "Dashboard > Connection Github kedua > Configuration". On the left, there is a sidebar with the title "Configure" and several menu items: "General", "Source Code Management", "Build Triggers" (which is highlighted), "Build Environment", "Build Steps", and "Post-build Actions". The main content area is for the "Build Triggers" section. It contains three checkboxes: "Trigger builds remotely (e.g., from scripts)" (unchecked), "Build after other projects are built" (unchecked), and "Build periodically" (checked). Below the "Build periodically" checkbox is a "Schedule" field with a help icon. The field contains the cron expression `*/* 2 * * *`. Below the field is a warning message: "Spread load evenly by using 'H/2 * * * *' rather than '*/* 2 * * *'". Below the warning is a text line: "Would last have run at Thursday, March 28, 2024 at 1:40:39 PM Coordinated Universal Time; would next run at Thursday, March 28, 2024 at 1:40:39 PM Coordinated Universal Time." At the bottom of the configuration area are two checkboxes: "GitHub hook trigger for GITScm polling" (unchecked) and "Poll SCM" (unchecked). At the very bottom are two buttons: "Save" and "Apply".

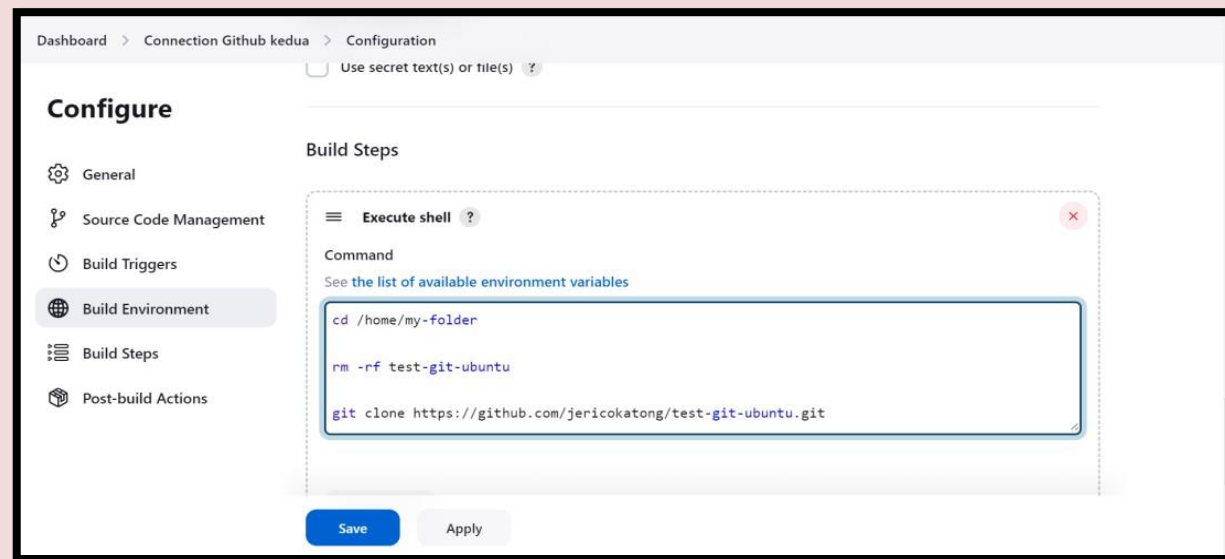
Note, jika pada bagian checkbox “github hook trigger....” masih kosong. Nanti bisa tambah install plugin github atau github integration di jenkins. Ini nanti bisa dipakai untuk trigger dari github.

1# Otomasi Git Clone dari Github

Opsi #1 – Otomasi menggunakan Git Clone.

Pada build step ini pilih execute shell, kemudian masukan script dibawah.

- 1) Penting, jangan lupa untuk membuat folder di **home** directory (*terserah*).
- 2) Disini telah dibuat sebuah folder di **home**, kemudian untuk make sure tidak ada folder **test-git-ubuntu** (*folder yang ada ketika lakukan git clone*) kita bisa pakai perintah **rm** untuk hapus jika ada folder tersebut.
- 3) Setelah itu, lakukan **git clone** dari link repo di github.
- 4) Ketiga command tersebut akan dijalankan oleh jenkins setiap 2 menit.
- 5) Setelah itu di save lagi !!!



Note: Kalian bisa langsung arahkan ke root directory dari project kalian:

- `cd /var/www/html/unkpresent`
- `rm -rf test-git-ubuntu`
- `git clone https://github.com/jericokatong/test-git-ubuntu.git`

2# Otomasi Git Pull dari Github

Opsi #2 – Otomasi menggunakan Git Pull.

Kita juga bisa menggunakan Git pull, perintah ini digunakan untuk **mengambil perubahan terbaru** dari **branch master** di repository remote (origin) dan menggabungkannya dengan branch local (di sisi server). Ini berguna untuk memastikan bahwa kode di server selalu diperbarui dengan versi terbaru dari repository Git.

The screenshot shows the Jenkins Configuration page for a job named 'connectUnkpresent'. The left sidebar contains a 'Configure' section with several options: General, Source Code Management, Triggers, Environment (which is currently selected), Build Steps, and Post-build Actions. The main area of the page is titled 'Build Steps' and includes a description: 'Automate your build process with ordered tasks like code compilation, testing, and deployment.' Below this, there is a dashed box containing a build step named 'Execute shell'. The 'Command' field for this step contains the following text: `cd /var/www/html/unkpresent`, `git add .`, and `git pull origin main`. At the bottom of the configuration page, there are two buttons: 'Save' and 'Apply'.

Dashboard > connectUnkpresent > Configuration

☐ Use secret text(s) or file(s) ?

Configure

- General
- Source Code Management
- Triggers
- Environment**
- Build Steps
- Post-build Actions

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

Execute shell ?

Command

[See the list of available environment variables](#)

```
cd /var/www/html/unkpresent  
  
git add .  
  
git pull origin main
```

Advanced ▾

Save **Apply**

[6]

Loading Forever pada Credential

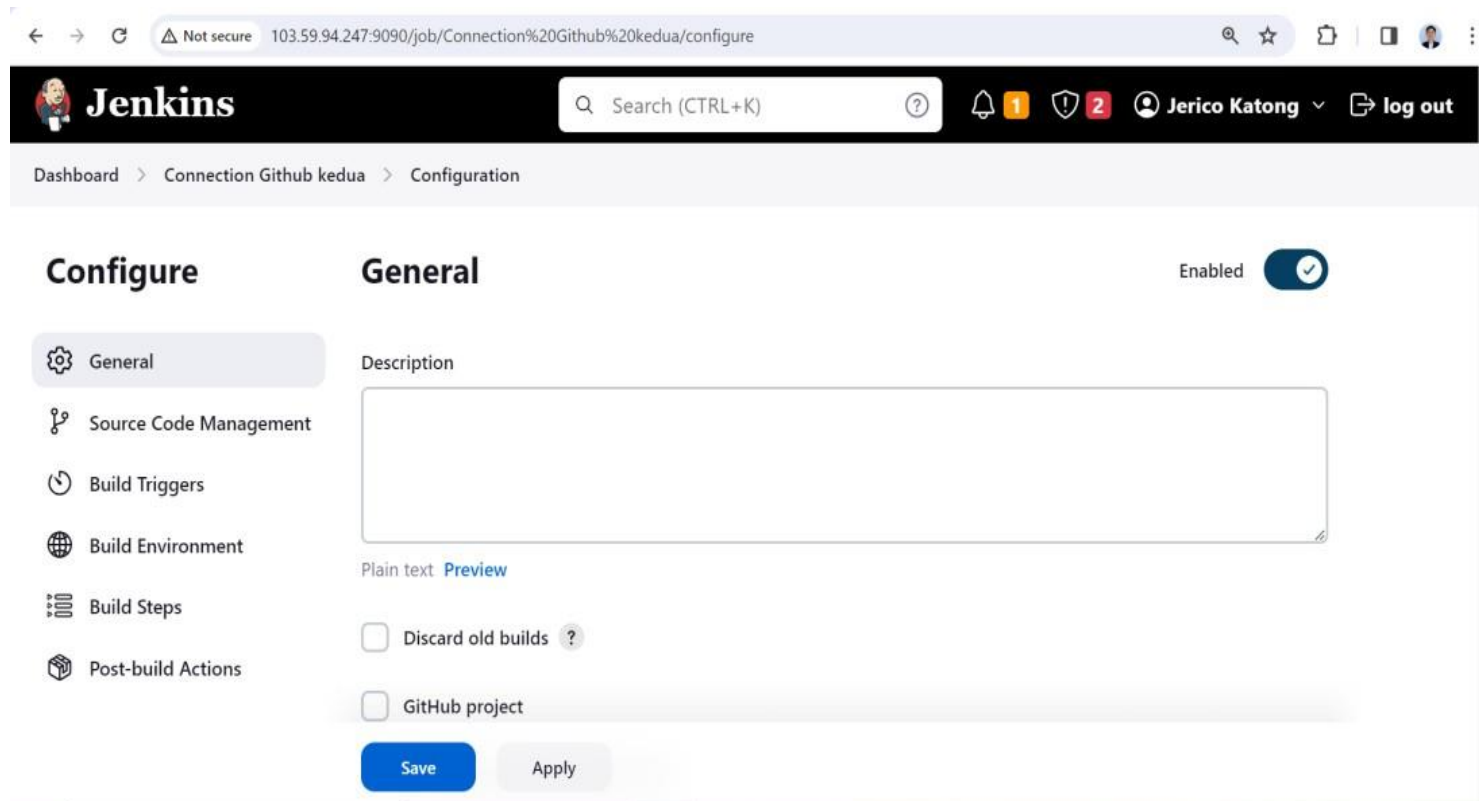


Jenkins

Loading Forever pada Credential

Salah satu alasan mengapa terjadi **loading forever** pada saat *load credential* yaitu mungkin karena keterbatasan memory (contoh karena RAM 2 GB saja), jika memiliki spec RAM VPS yang lebih dari 4 GB harusnya tidak bermasalah.

Step #1: Cek Kembali isi description kemudian **save**.

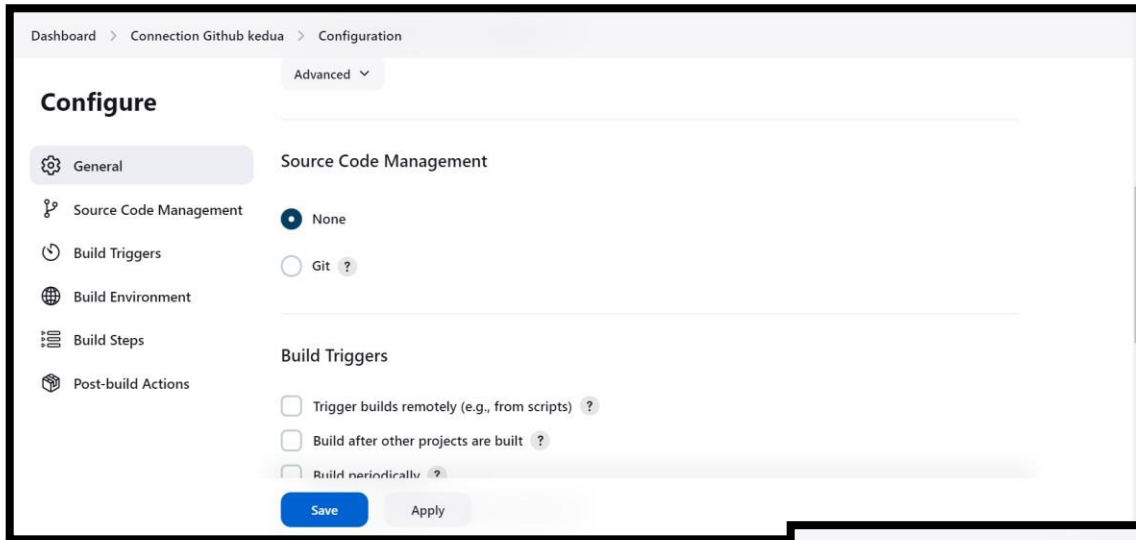


The screenshot shows the Jenkins web interface. The browser address bar indicates the URL: `103.59.94.247:9090/job/Connection%20Github%20kedua/configure`. The Jenkins header includes the logo, a search bar, and user information for "Jerico Katong". The breadcrumb trail shows: Dashboard > Connection Github kedua > Configuration.

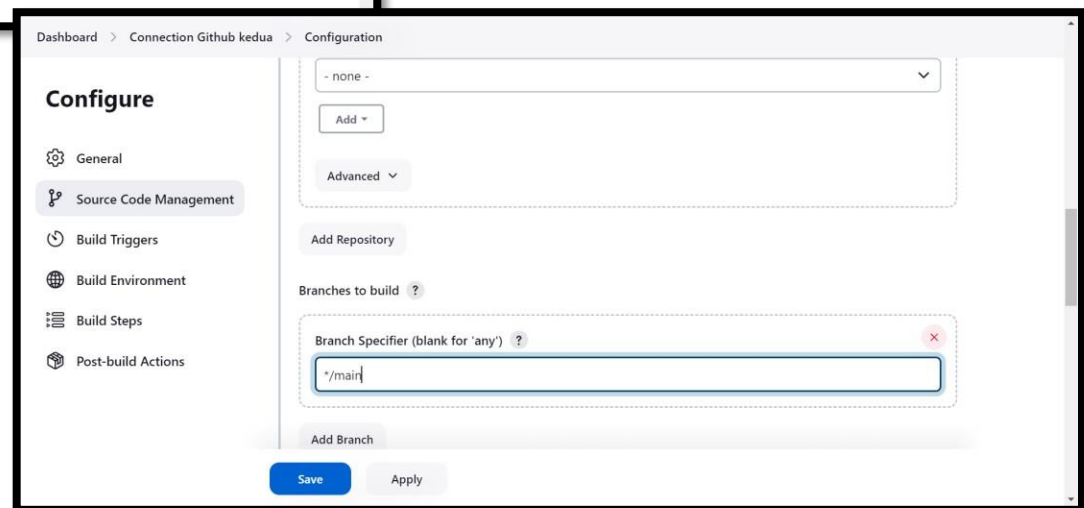
The "Configure" section is active, showing a sidebar with options: General (selected), Source Code Management, Build Triggers, Build Environment, Build Steps, and Post-build Actions. The "General" tab is selected, and the "Description" field is empty. Below the description field, there are two checkboxes: "Discard old builds" and "GitHub project", both of which are unchecked. At the bottom, there are "Save" and "Apply" buttons. The "Enabled" toggle switch is turned on.

Configure Job Jenkins Kembali

Step #2: Setelah itu klik **configure** kembali. Kemudian di Source Code Management klik Git, kemudian save lagi.

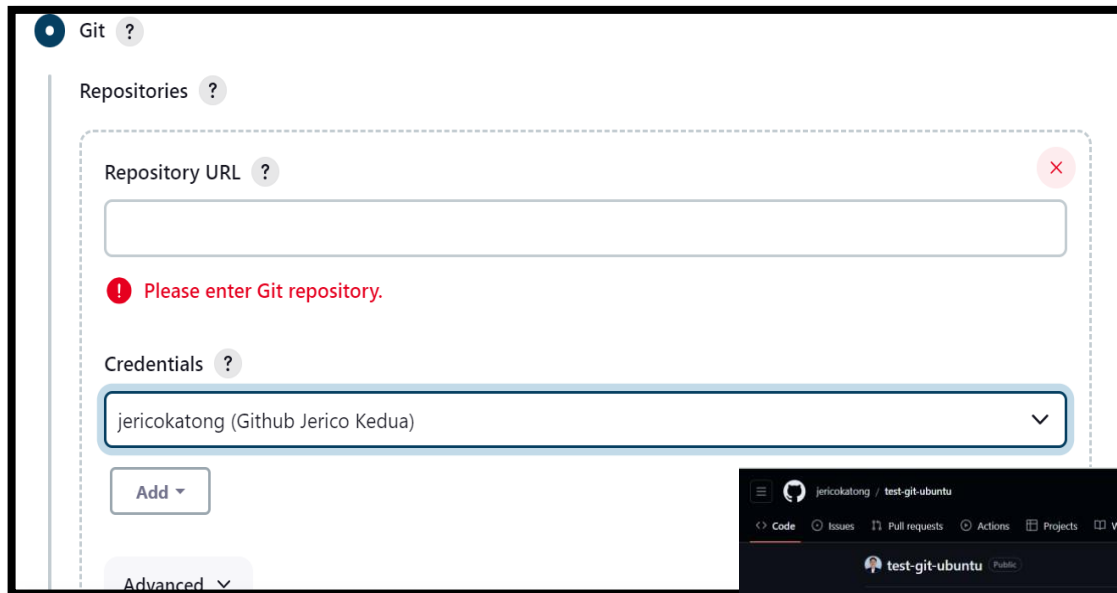


Step #3: Configure lagi, kemudian masukan branch, disini branch hanya main saja. Kemudian Save lagi.



Tips & Tricks

Setelah itu, balik lagi ke configure, balik ke bagian credential, selanjutnya pilih credential yg sudah kita masukan. **Tapi IMPORTANT THING, kita pilih credential terlebih dahulu !!!**. Setelah itu save lagi.



Git ?

Repositories ?

Repository URL ?

Please enter Git repository.

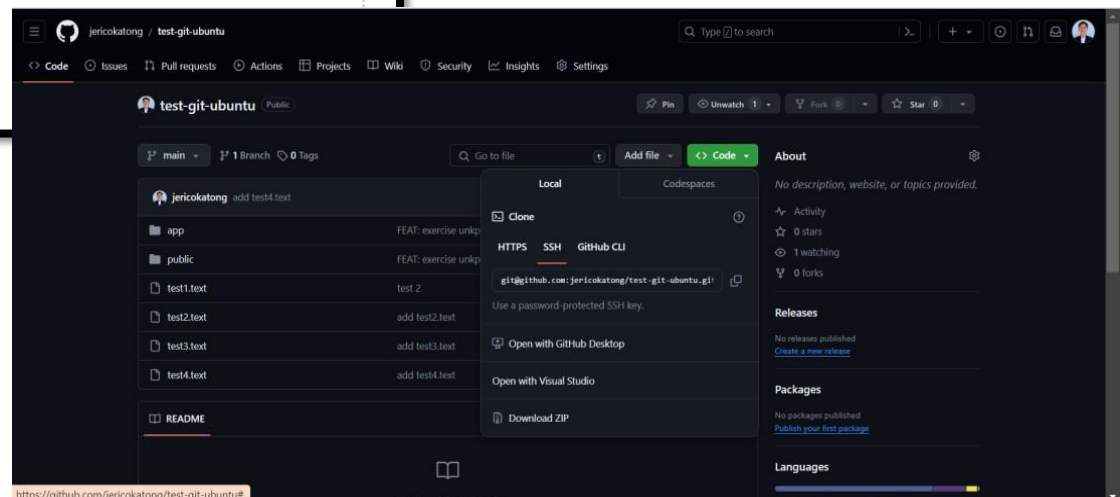
Credentials ?

jericokatong (Github Jerico Kedua)

Add ▾

Advanced ▾

Setelah itu masukan link *ssh* dari github ke repo url Github.

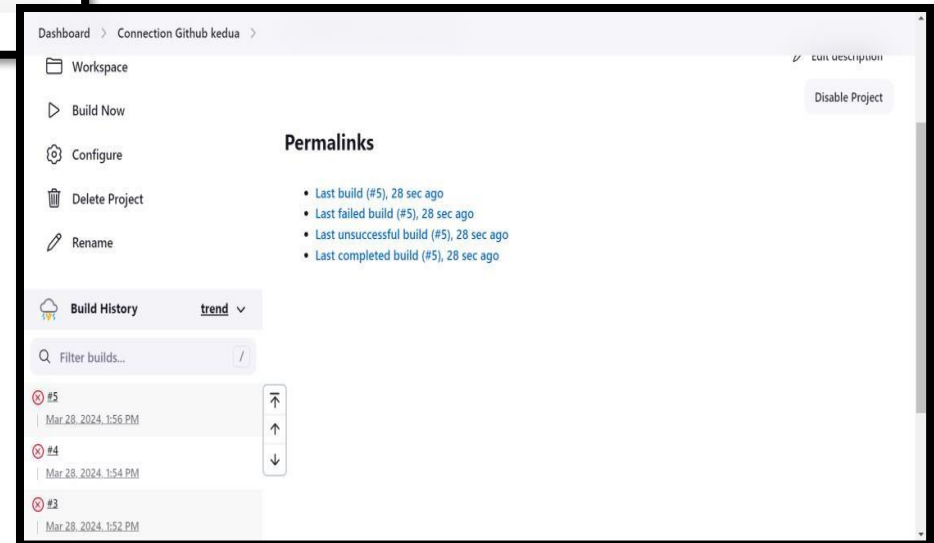


Tips & Tricks

Jika sudah tidak ada lagi “loading forever”. Maka save lagi.

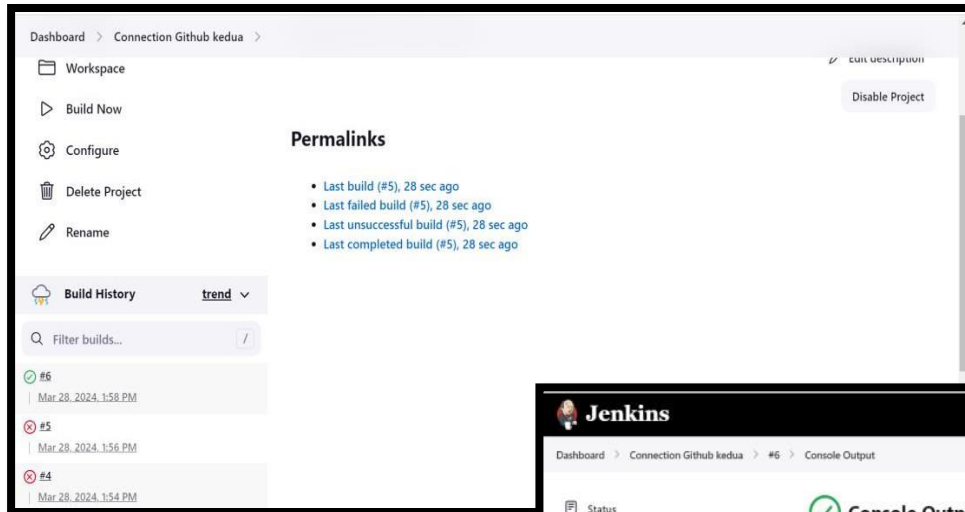


Selanjutnya, kita tunggu proses otomasi build dari Jenkins Job. Pada bagian Build History, akan muncul riwayat prosesnya downloadnya, yang akan di clone secara otomatis dari Github. Warna merah menandakan proses sesuai. Ditunggu saja ya sampai warna berubah.

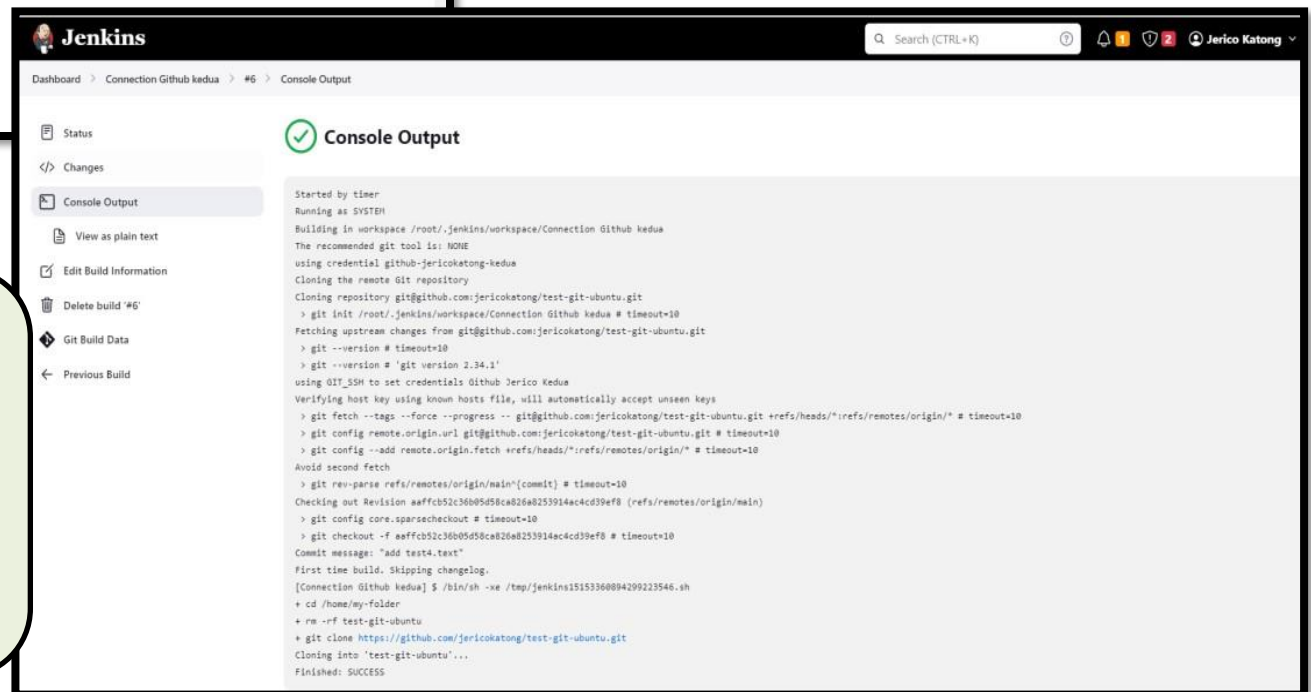


Tips & Tricks

Ini contoh yang sudah berhasil di clone, warna build historynya jadi warna hijau.



Bisa dilihat juga pada bagian console outputnya. Sudah jadi script otomasi git clone yang telah dibuat.



Kill Jenkins Service (Java)

Very high CPU usage, sometimes over/reach to 100% (there are only 2 cores).

htop

```
1 [|||||199.4%] Tasks: 100, 2201 thr; 11 running
2 [|||||196.6%] Load average: 9.56 7.50 6.21
3 [|||||100.0%] Uptime: 09:17:53
4 [|||||100.0%]
Mem[|||||5441/7979MB]
Swp[|||||0/1023MB]

PID USER      PRI  NI  VIRT   RES   SHR  S CPU% MEM%   TIME+  Command
13276 jenkins    20   0 4525M 2229M 7820  S 113. 27.9 23h31:28 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/jenkins/jenkins.war --web
25861 root        20   0 458M  117M 55156  R 97.7  1.5  2:43.81 /usr/bin/php /lib/vendor/drush/drush/drush.php --php=/usr/bin/php --php-options=
25658 root        20   0 455M  114M 52684  R 96.6  1.4  3:42.81 /usr/bin/php /lib/vendor/drush/drush/drush.php --php=/usr/bin/php --php-options=
17444 jenkins    20   0 4525M 2229M 7820  R 31.2 27.9 5h42:42 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/jenkins/jenkins.war --web
27479 jenkins    20   0 400M  54580 38668  R 26.7  0.7  0:04.29 /usr/bin/php /lib/vendor/drush/drush/drush.php --php=/usr/bin/php --php-options=
17442 jenkins    20   0 4525M 2229M 7820  R 25.6 27.9 5h42:26 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/jenkins/jenkins.war --web
17443 jenkins    20   0 4525M 2229M 7820  R 25.0 27.9 5h42:33 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/jenkins/jenkins.war --web
17441 jenkins    20   0 4525M 2229M 7820  R 24.4 27.9 5h42:20 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/jenkins/jenkins.war --web
27680 root        20   0 27784  3832 1412  R  8.5  0.0  0:00.78 htop
1267 mysql      20   0 1718M 716M 4944  S  8.5  9.0 12:50.77 /usr/sbin/mysqld
18135 mysql      20   0 1718M 716M 4944  S  7.4  9.0  1:29.98 /usr/sbin/mysqld
17445 jenkins    20   0 4525M 2229M 7820  S  6.8 27.9 14:10.65 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/jenkins/jenkins.war --web
27638 jenkins    20   0 85140  5516 2920  R  2.8  0.1  0:00.26 git -c core.askpass=true fetch --tags --progress git@github.com:NBC-Sports-Group
4206 mysql      20   0 1718M 716M 4944  S  1.1  9.0  0:40.39 /usr/sbin/mysqld
4371 root        20   0 180M  7740 1748  S  0.6  0.1  2:12.84 /usr/bin/python /usr/bin/fail2ban-server -b -s /var/run/fail2ban/fail2ban.sock -
7798 nobody     20   0 15.9G  244M 1644  S  0.6  3.1  0:04.16 /usr/sbin/varnishd -P /var/run/varnishd.pid -a :80 -T localhost:6082 -f /etc/var
10472 jenkins    20   0 4550M 856M 9324  S  0.6 10.7 0:00.20 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/jenkins/jenkins.war --web
27642 jenkins    20   0 4550M 856M 9324  R  0.6 10.7 0:00.01 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/jenkins/jenkins.war --web
10362 jenkins    20   0 4550M 856M 9324  S  0.0 10.7 3:51.65 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/jenkins/jenkins.war --web
F1Help F2Setup F3Search F4Filter F5Tree F6SortBy F7Nice F8Nice F9Kill F10Quit
```

How to fix it???, try to delete/kill most of the old builds from the **problem jobs** using this strategy and then reload jenkins.

- sudo su
- htop
- ps -ef| grep jenkins # Cara untuk mendapatkan PID
- sudo kill -9 **13276**

Remove Jenkins Data

Sebelum menghapus data, pastikan Jenkins tidak berjalan.

- **sudo systemctl stop jenkins**
- **sudo kill -9 13276** #(bisa juga menggunakan PID, opsional)

Secara default, data Jenkins biasanya disimpan di direktori *jenkins_home*. Directory ini berisi semua data, termasuk konfigurasi, plugin, build, dan log. Kita dapat menghapus folder ini dengan command:

- **sudo rm -rf /var/lib/jenkins**

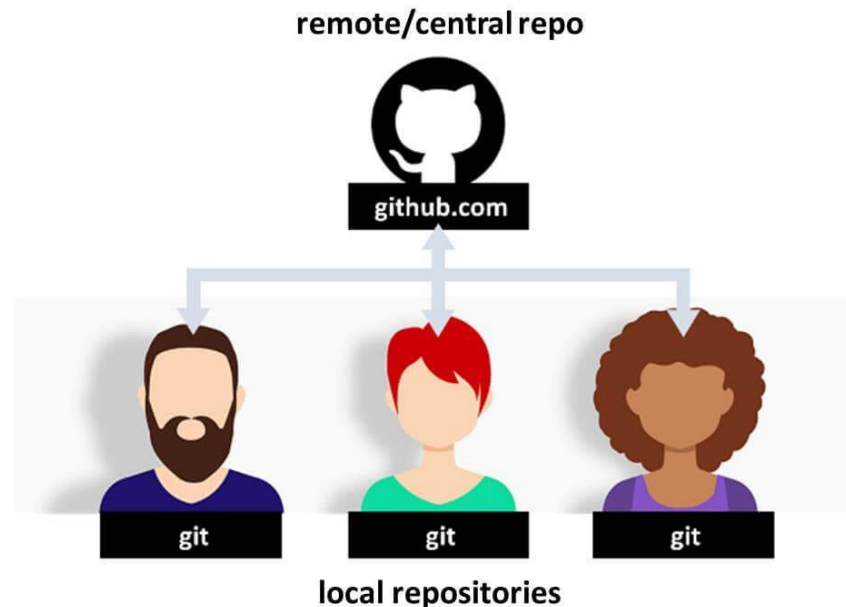
Hapus file konfigurasi (opsional):

- **sudo rm -rf /etc/jenkins** # Hapus konfigurasi
- **sudo rm -rf /var/log/jenkins** # Hapus log Jenkins
- **sudo rm -rf /var/cache/jenkins** # Hapus cache Jenkins

Note:

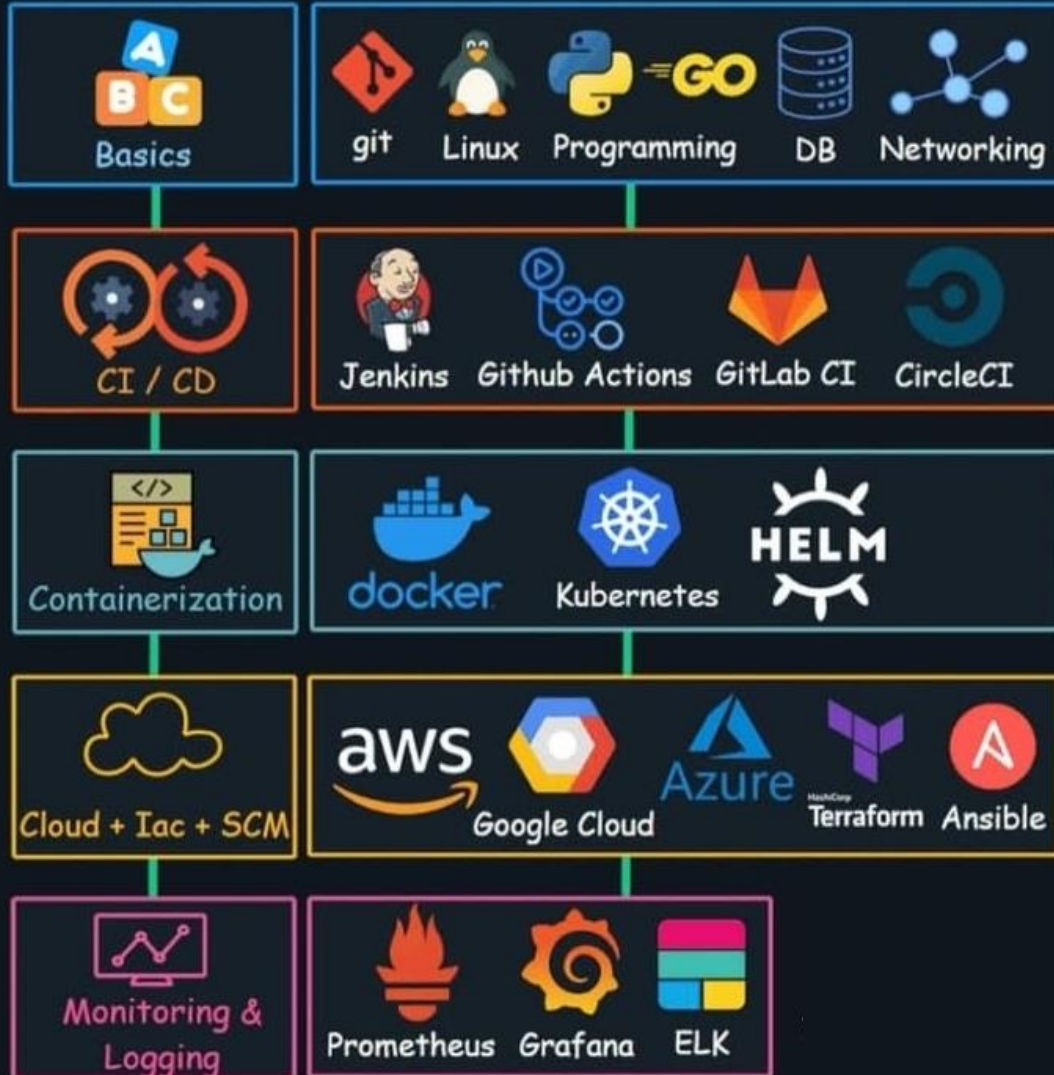
Semua data dan file Jenkins seharusnya sudah terhapus dari sistem, termasuk Git plugin yang kalian install sebelumnya. Jadi mohon hati-hati sebelum mengikuti langkah ini.

- 1) Berdasarkan pengalaman di semester sebelumnya, untuk demo class activity ini RAM server untuk menjalankan Jenkins yaitu ≥ 4 GB. Jika memungkinkan stop dulu beberapa service, diantaranya MySQL. Silahkan resize server masing-masing kelompok.
- 2) Berapa hal juga yang bisa menjadi sumber masalah adalah **SSH key public dan private**. Coba hapus SSH key public dan private dan buat kembali dan username github, email dan email github yang terhubung di server.



DevOps Learning Path

For **Beginners**



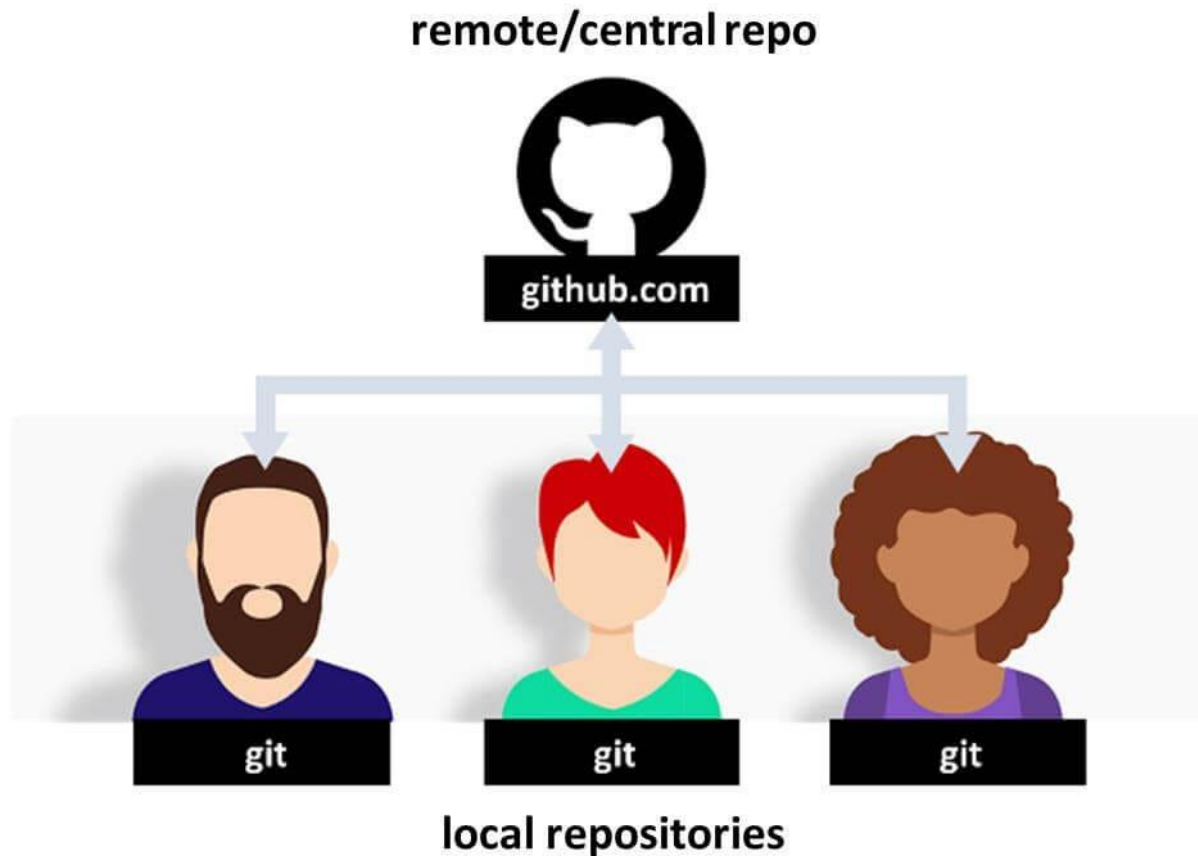
Typical SCM process:

(React, Vue, Svelte, atau Angular)

- 1) Developer **clones** a Git repository.
- 2) Makes changes to code.
- 3) **Commits** the changes locally with a message.
- 4) **Pushes** changes to a remote repository (e.g., GitHub).
- 5) A **Pull Request (PR)** or **Merge Request (MR)** is reviewed by teammates.
- 6) Once approved, changes are merged into the main branch.
- 7) The CI/CD pipeline (e.g., Jenkins, GitHub Actions) automatically builds and deploys the new version.

Framework Frontend & Backend

Framework / Teknologi	Dibuat oleh	Jenis	Bahasa Dasar	Kelebihan Utama
React	Meta (Facebook)	Library Frontend	JavaScript	Populer, ekosistem besar, komponen reusable
Vue.js	Evan You	Framework Frontend	JavaScript	Mudah dipelajari, dokumentasi jelas, sintaks sederhana
Svelte	Rich Harris	Compiler Frontend	JavaScript	Cepat, kode bersih, ringan tanpa virtual DOM
Angular	Google	Full Frontend Framework	TypeScript	Lengkap (routing, DI, form, HTTP), enterprise-ready
Django	Django Software Foundation	Full Backend Framework	Python	Cepat dikembangkan, ORM kuat, keamanan tinggi
Laravel	Taylor Otwell	Full Backend Framework	PHP	Elegant, sintaks modern, banyak fitur bawaan (Auth, Queue, API)
CodeIgniter (CI)	EllisLab / BCIT	Lightweight Backend Framework	PHP	Ringan, mudah dipahami, cocok untuk pemula
PHP Native	—	Bahasa Server-side	PHP	Tanpa framework, fleksibel, langsung ke server



Next meeting, kalian akan diminta untuk Clone/Pull git remote repository secara otomatis dengan menggunakan Jenkins. **Single/Multiple user** yang akan push code ke one github repo dan Jenkins akan clone/pull github repo tersebut ke server. Bahasa pemrograman tidak di batasi, gunakan Bahasa pemrograman (front-end / back-end) yang kalian sukai.

Buat project sederhana,

Tidak di Batasi untuk Bahasa pemrograman tertentu.
Demo CI/CD di dalam class per kelompok.

- (1) Create Job Jenkins.
- (2) Edit code dan push dari local.
- (3) Git Clone/pull (*mengambil perubahan terbaru*) secara otomatis.
- (4) Connect ke database.
- (5) CRUD functions.
- (6) Buat PPT presentasi per kelompok (*dijelaskan*).

Jika kalian menganggap Jenkins ini buat apa, tidak ada gunanya.
Maka itu salah !!!

Jenkins adalah *powerful automation tool* yang dapat digunakan untuk berbagai jenis *projects* dan *tasks* dalam **software development** dan **system administration**.

END PRESENTATION

Thank you for your attention

Instructor: S – W – T

THANK YOU

